Asymptotic center–manifold for systems with parameter perturbations and harmonic forcings

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Abstract

Asymptotic center-manifold approximations are constructed for systems that have been perturbed slightly away from the bifurcation point in the parameter space. This is done by extending the existing system to include parameter perturbations as dynamic variables, leading to an extended critical subspace and dimensionality of the center-manifold. The extension is done in a very general setting and it is shown that the consequences of such a system extension are non-trivial. The extended critical subspace has a certain structure associated with it, which may be deduced from the original system. The application of the center-manifold theorem to the extended system leads to a graph equation which rigorously valid in the vicinity of the bifurcation point (even in parameter space). The graph equation is approximated as an asymptotic power series in the amplitudes of the critical eigenmodes leading to general expressions for restricted resolvent solutions for the various terms in the power series at each order. A Ginzburg-Landau model near bifurcation is used to demonstrate the validity of the asymptotic center-manifold in the extended space.

1 Problem Definition

1.1 The Standard System

Consider a system with its state variable denoted by $\boldsymbol{u}$, consisting of l parameters given by the vector $\boldsymbol{\mu}$, and an evolution law given by

$$\frac{\partial \boldsymbol{u}}{\partial t} = \boldsymbol{f}(\boldsymbol{u}, \boldsymbol{\mu}). \tag{1}$$

The system state $\boldsymbol{u}$ is N-dimensional where, N is assumed to be large. The evolution law may arise from a finite set of ordinary differential equations or, may be a finite representation of an infinite dimensional partial differential equation as often arises from the numerical discretization of the space-time evolution equations of physical systems. Typically, physical systems will have real parameters $\boldsymbol{\mu}$ however, allowing for complex parameters presents no additional difficulty and we assume that $\boldsymbol{\mu} \in \mathbb{C}^l$. Without loss of generality one may assume $\boldsymbol{u} = \boldsymbol{u}^b$ to be a fixed point of the system for a certain set of parameters $\boldsymbol{\mu} = \boldsymbol{\mu}^b$, *i.e.* ,

$$\boldsymbol{f}(\boldsymbol{u}^b, \boldsymbol{\mu}^b) = \mathbf{0}. \tag{2}$$

We may consider the evolution of the deviation from the fixed point solution $\mathbf{u} = \boldsymbol{u} - \boldsymbol{u}^b$, and divide the evolution operator into its linear and non-linear parts as,

$$\frac{\partial \mathbf{u}}{\partial t} = \mathbf{L}\mathbf{u} + \mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b) \tag{3}$$

where,

$$\mathbf{L} = \left[\left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_{(\boldsymbol{u}=\boldsymbol{u}^b, \boldsymbol{\mu}=\boldsymbol{\mu}^b)} \right]$$
$$\mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b) = \boldsymbol{f}(\boldsymbol{u}^b + \mathbf{u}, \boldsymbol{\mu}^b) - \mathbf{L}\mathbf{u}.$$

Furthermore, it is assumed that at $\boldsymbol{\mu} = \boldsymbol{\mu}^b$ the system undergoes a multiple co-dimension bufurcation and that the linearized operator $\mathbf{L} \in \mathbb{C}^{N \times N}$ has a discrete set of n critical eigenvalues $\lambda_i^C,\ i \in 1, \ldots, n$, with zero growth rate ($\mathfrak{Re}(\lambda_i^C) = 0$). This includes both purely real critical eigenvalues with $\mathfrak{Re}(\lambda_i^C) = 0,\ \mathfrak{Im}(\lambda_i^C) = 0$, and purely imaginary critical eigenvalues with $\mathfrak{Re}(\lambda_i^C) = 0,\ \mathfrak{Im}(\lambda_i^C) \neq 0$. Accordingly, the system has n critical eigenpairs $(\lambda_i^C, \mathbf{v}_i), i \in 1 \ldots n$, where $\mathbf{v}_i$ represents the eigenvector corresponding to λ_i^C. For real systems the purely imaginary critical eigenvalues will occur in conjugate pairs.

The adjoint of the direct linearized operator $\mathbf{L}$ is denoted as $\mathbf{L}^\dagger$ which also has n critical eigenpairs $(\lambda_i^{C*}, \mathbf{w}_i)$. The scaling of the eigenvectors of the direct and adjoint operators may be defined such that they satisfy the birthogonality condition

$$\langle \mathbf{w}_i, \mathbf{v}_j \rangle = \mathbf{w}_i^\dagger \mathbf{v}_j = \delta_{i,j}, \tag{4}$$

where $\langle \cdot, \cdot \rangle$ is the standard inner-product and $\delta_{i,j}$ is the Kronecker-delta fuction. The superscript † is used here to indicate a complex-conjugated transpose of vectors and matrix operators. For scalars complex-conjugation is indicated by a * superscript. The superscript ^{H} which is often used for the same purpose is reserved for objects referring to harmonic forcings. One may define the matrices $\mathbf{V}$ and $\mathbf{W}$, whose columns comprise of the n critical eigenvectors of $\mathbf{L}$ and $\mathbf{L}^\dagger$as,

$$\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n], \tag{5a}$$
$$\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n]. \tag{5b}$$

Due to the chosen scaling denoted by equation (4) the matrices satisfy,

$$\mathbf{W}^\dagger \mathbf{V} = \mathbf{I}_n, \tag{6}$$

where, $\mathbf{I}_n$ represents an identity matrix of order $n \times n$. The columns of the matrices $\mathbf{V}$ and $\mathbf{W}$ span the n-dimensional center-subspace of the linear operators $\mathbf{L}$ and $\mathbf{L}^\dagger$ respectively.

Additionally, $\boldsymbol{\Lambda}_C$ is defined as an $n \times n$ diagonal matrix consisting of the critical eigenvalues of $\mathbf{L}$, with the implicit assumption of diagonalizability of $\mathbf{L}$ within the center subspace. The ordering of the eigenvalues in $\boldsymbol{\Lambda}_C$ and eigenvectors in $\mathbf{V}$ and $\mathbf{W}$ is consistent with each other so that we have the eigenvalue relations,

$$\mathbf{L}\mathbf{V} = \mathbf{V}\boldsymbol{\Lambda}_C, \qquad \mathbf{L}^\dagger \mathbf{W} = \mathbf{W}\boldsymbol{\Lambda}_C^\dagger. \tag{7}$$

1.2 From the Standard System to the Extended System

The same system is now considered for the case of parameter perturbation so that the evolution dynamics are now evaluated at parameter values of $\boldsymbol{\mu} = \boldsymbol{\mu}^b + \boldsymbol{\nu}$, with $\boldsymbol{\nu}$ being the vector of parameter perturbations, $\boldsymbol{\nu} = [\nu_1; \nu_2; \ldots; \nu_l]$. In addition, the system is forced with harmonic forcing of the type

$$\boldsymbol{f}^H(\boldsymbol{\theta}) = \sum_{i=1}^{h} \boldsymbol{f}_i^H \theta_i(t); \quad \partial\theta_i/\partial t = \lambda_i^H \theta_i, \tag{8}$$

where, $\boldsymbol{\theta} = [\theta_1; \theta_2; \ldots; \theta_h]$ and the various θ_i are treated as dynamic variables. λ_i^H is purely imaginary, with the imaginary part representing the angular frequency of the forcing. The vector of angular frequencies is denoted as $\boldsymbol{\lambda}^H = [\lambda_1^H; \lambda_2^H; \ldots; \lambda_h^H]$. Constant forcing terms are readily incorporated when a particular (or multiple) $\lambda_i^H = 0$. The (time-independent) shape of the i^{th} harmonic forcing is denoted as $\boldsymbol{f}_i^H$. Denoting the new harmonically forced evolution operator as,

$$\boldsymbol{f}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta}) = \boldsymbol{f}(\boldsymbol{u}^b + \mathbf{u}, \boldsymbol{\mu}^b + \boldsymbol{\nu}) + \boldsymbol{f}^H(\boldsymbol{\theta}), \tag{9}$$

one may again separate out the linear and nonlinear parts as,

$$\mathbf{L} = \left[\left.\frac{\partial \boldsymbol{f}'}{\partial \mathbf{u}}\right|_{(\mathbf{u}=\mathbf{0}, \boldsymbol{\nu}=\mathbf{0}, \boldsymbol{\theta}=\mathbf{0})} \right] = \left[\left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_{(\boldsymbol{u}=\boldsymbol{u}^b, \boldsymbol{\mu}=\boldsymbol{\mu}^b)} \right],$$
$$\mathbf{L}_{\boldsymbol{\nu}} = \left[\left.\frac{\partial \boldsymbol{f}'}{\partial \boldsymbol{\nu}}\right|_{(\mathbf{u}=\mathbf{0}, \boldsymbol{\nu}=\mathbf{0}, \boldsymbol{\theta}=\mathbf{0})} \right] = \left[\left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{\mu}}\right|_{(\boldsymbol{u}=\boldsymbol{u}^b, \boldsymbol{\mu}=\boldsymbol{\mu}^b)} \right],$$
$$\mathbf{L}_{\boldsymbol{\theta}} = \left[\left.\frac{\partial \boldsymbol{f}'}{\partial \boldsymbol{\theta}}\right|_{(\mathbf{u}=\mathbf{0}, \boldsymbol{\nu}=\mathbf{0}, \boldsymbol{\theta}=\mathbf{0})} \right] = \left[\left.\frac{\partial \boldsymbol{f}^H}{\partial \boldsymbol{\theta}}\right|_{(\boldsymbol{\theta}=\mathbf{0})} \right],$$

$$\mathcal{N}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta}) = \boldsymbol{f}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta}) - \mathbf{L}\mathbf{u} - \mathbf{L}_{\boldsymbol{\nu}}\boldsymbol{\nu} - \mathbf{L}_{\boldsymbol{\theta}}\boldsymbol{\theta}.$$

where, $\mathbf{L} \in \mathbb{C}^{N\times N}$, $\mathbf{L}_{\boldsymbol{\nu}} \in \mathbb{C}^{N\times l}$, $\mathbf{L}_{\boldsymbol{\theta}} \in \mathbb{C}^{N\times h}$ which results in the evolution equation given by,

$$\frac{\partial \mathbf{u}}{\partial t} = \mathbf{L}\mathbf{u} + \mathbf{L}_{\boldsymbol{\nu}}\boldsymbol{\nu} + \mathbf{L}_{\boldsymbol{\theta}}\boldsymbol{\theta} + \mathcal{N}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta}).$$

The diagonal matrix for the harmonic angular frequencies is represented by $\boldsymbol{\Lambda}_H = diagm(\boldsymbol{\lambda}^H)$. Similarly, $\boldsymbol{\lambda}^P = [\lambda_1^P; \lambda_2^P; \ldots; \lambda_l^P]$ is defined as a vector of length l, with all $\lambda_i^P = 0$ identically, for $i \in 1, \ldots, l$. These may be thought of parameter eigenvalues or parameter angular frequencies depending on the point of view. In either case, these vanish identically. $\boldsymbol{\Lambda}_P = diagm(\boldsymbol{\lambda}^P)$ is defined as an $l \times l$ null matrix, to represent the trivial linear evolution matrix for the parameter variables $\boldsymbol{\nu}$. The extended system including the evolution equations for the newly introduced variables $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$, henceforth referred to as the *extended variables*, is given by,

$$\frac{\partial \mathbf{u}}{\partial t} = \mathbf{L}\mathbf{u} + \mathbf{L}_{\boldsymbol{\nu}}\boldsymbol{\nu} + \mathbf{L}_{\boldsymbol{\theta}}\boldsymbol{\theta} + \mathcal{N}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta}) \tag{10a}$$

$$\frac{d\boldsymbol{\nu}}{dt} = \boldsymbol{\Lambda}_P \boldsymbol{\nu} \tag{10b}$$

$$\frac{d\boldsymbol{\theta}}{dt} = \boldsymbol{\Lambda}_H \boldsymbol{\theta}, \tag{10c}$$

An overhead hat notation, $\widehat{\cdot}$, is used to refer to variables and operators associated with the extend space, so that the new state vector, non-linear term and the linearized direct and adjoint operators are denoted by $\widehat{\boldsymbol{u}}$, $\widehat{\mathcal{N}'}$, $\widehat{\mathbf{L}}$ and $\widehat{\mathbf{L}}^\dagger$, that are defined as,

$$\widehat{\mathbf{u}} = [\mathbf{u}; \boldsymbol{\nu}; \boldsymbol{\theta}], \qquad \widehat{\mathcal{N}'} = [\mathcal{N}'; \mathbf{0}; \mathbf{0}],$$

$$\widehat{\mathbf{L}} = \begin{bmatrix} \mathbf{L} & \mathbf{L}_{\boldsymbol{\nu}} & \mathbf{L}_{\boldsymbol{\theta}} \\ \mathbf{0} & \boldsymbol{\Lambda}_P & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{\Lambda}_H \end{bmatrix}, \quad \widehat{\mathbf{L}}^\dagger = \begin{bmatrix} \mathbf{L}^\dagger & \mathbf{0} & \mathbf{0} \\ \mathbf{L}_{\boldsymbol{\nu}}^\dagger & \boldsymbol{\Lambda}_P^\dagger & \mathbf{0} \\ \mathbf{L}_{\boldsymbol{\theta}}^\dagger & \mathbf{0} & \boldsymbol{\Lambda}_H^\dagger \end{bmatrix}. \tag{11}$$

The full system is now M-dimensional, with $M = N + l + h$, and may be represented compactly as,

$$\frac{\partial \widehat{\mathbf{u}}}{\partial t} = \widehat{\mathbf{L}}\widehat{\mathbf{u}} + \widehat{\mathcal{N}'}(\widehat{\mathbf{u}}). \tag{12}$$

The system defined by equation (10), or its compact form in equation (12), with additional dynamical equations for the extended variables will be referred to as the *extended system*.

2 Extended Subspace Structure

The extended system has a larger center subspace of dimension $m = n + l + h$, as compared to the original system due to the $l+h$ additional variables introduced. While the extension of the system seems almost trivial, the eigenvector structure in the new dimensions however is far from trivial. Nevertheless, the extended eigenspace structure can be deduced directly from the properties of the original system. It is useful to examine the structure of the extended subspace since the the adjoint extended critical subspace does have a special structure which has important practical implications for the asymptotic evaluation of the center-manifold of the extended system.

There are however some subtleties involved when the system extension introduces new eigenvalues that are resonant with the original linear system $\mathbf{L}$. In such a scenario the system extension leads to a generalized eigenspace for the resonant eigenvalues. The subspace structure is now evaluated in detail for both the resonant and non-resonant cases. This is first done for the extended operator $\widehat{\mathbf{L}}$ and then for the adjoint operator $\widehat{\mathbf{L}}^\dagger$.

2.1 Subspace structure for $\widehat{\mathbf{L}}$

Observe that if $(\lambda_i^C, \mathbf{v}_i)$ is a critical eigenpair of the original linearized operator $\mathbf{L}$, then its extension to $\widehat{\mathbf{L}}$ satisfies,

$$\widehat{\mathbf{v}}_i^C = [\mathbf{v}_i; \mathbf{0}; \mathbf{0}], \tag{13a}$$

$$(\lambda_i^C \widehat{\mathbf{I}} - \widehat{\mathbf{L}})\widehat{\mathbf{v}}_i^C = \mathbf{0}, \tag{13b}$$

$\forall i \in 1, \ldots, n$, which may be verified via simple substitution. Here $\widehat{\mathbf{I}}$ is the identity matrix for the extended system. Hence all critical eigenmodes of $\mathbf{L}$ can be trivially extended to $\widehat{\mathbf{L}}$. This definition for the extension of the eigenvectors in fact applies to *all* eigenmodes of $\mathbf{L}$.

The additional critical eigenpairs introduced due to system extension need to be deduced. The new eigenmodes introduced due to parameter perturbations will be referred to as the *parameter* modes while those generated due to the harmonic forcings will be referred to as *forcing* modes. Accordingly, two sets of canonical unit vectors are defined such that,

$$\boldsymbol{e}_i^P = [\overbrace{0;0;\ldots;0}^{i-1};1;\overbrace{0;0;\ldots;0}^{l-i}], \tag{14a}$$
$$\boldsymbol{e}_i^H = [\underbrace{0;0;\ldots;0}_{i-1};1;\underbrace{0;0;\ldots;0}_{h-i}]. \tag{14b}$$

If the original linear system $\mathbf{L}$ is non-singular then, the strcuture of the l parameter eigenpairs is given by $(\lambda_i^P, \widehat{\mathbf{v}}_i^P)$, which satisfy

$$\widehat{\mathbf{v}}_i^P = [\mathbf{v}_i^P; \boldsymbol{e}_i^P; \mathbf{0}], \quad \forall i \in 1, \ldots l, \tag{15a}$$
$$(\lambda_i^P \mathbf{I} - \mathbf{L})\mathbf{v}_i^P = \mathbf{L}_{\boldsymbol{\nu}} \boldsymbol{e}_i^P, \qquad (\lambda_i^P = 0). \tag{15b}$$

It can be verified that $(\lambda_i^P \widehat{\mathbf{I}} - \widehat{\mathbf{L}})\widehat{\mathbf{v}}_i^P = 0$.

However, if $\mathbf{L}$ is singular then equation (15) is indeterminate and the definition needs to be modified. For the i^{th} parameter mode, define a vector $\boldsymbol{\gamma}_i^P \in \mathbb{C}^n$, where the j^{th} element of the vector is given by,

$$(\boldsymbol{\gamma}_i^P)_j = \begin{cases} \mathbf{w}_j^\dagger \mathbf{L}_{\boldsymbol{\nu}} \boldsymbol{e}_i^P, & \textit{if } \lambda_i^P = \lambda_j^C, \\ 0, & \textit{otherwise}, \end{cases} \tag{16}$$

and we have a new definition of $\widehat{\mathbf{v}}_i^P$ satisfying,

$$\widehat{\mathbf{v}}_i^P = [\mathbf{v}_i^P; \boldsymbol{e}_i^P; \mathbf{0}], \quad \forall i \in 1, \ldots l, \tag{17a}$$
$$(\lambda_i^P \mathbf{I} - \mathbf{L})\mathbf{v}_i^P = \mathbf{L}_{\boldsymbol{\nu}} \boldsymbol{e}_i^P - \mathbf{V}\boldsymbol{\gamma}_i^P, \qquad (\lambda_i^P = 0), \tag{17b}$$
$$\mathbf{w}_j^\dagger \mathbf{v}_i^P = 0, \quad \textit{if } \lambda_i^P = \lambda_j^C. \tag{17c}$$

In the singular case, $\widehat{\mathbf{v}}_i^P$ is no longer an eigenvector of $\widehat{\mathbf{L}}$, however, it is a generalized eigenvector satisfying $(\lambda_i^P \widehat{\mathbf{I}} - \widehat{\mathbf{L}})^2 \widehat{\mathbf{v}}_i^P = \mathbf{0}$. Note that $\boldsymbol{\gamma}_i^P$ is identically zero for the non-singular case and equation (17) reduces to equation (15). Henceforth the parameter modes will be defined by equation (17), since it is the more general definition.

Similarly, when the harmonic forcing is non-resonant, the h forcing modes generated by system extension due to $\boldsymbol{\theta}$ have the following structure,

$$\widehat{\mathbf{v}}_i^H = [\mathbf{v}_i^H; \mathbf{0}; \boldsymbol{e}_i^H], \quad \forall i \in 1, \ldots h, \tag{18a}$$
$$(\lambda_i^H \mathbf{I} - \mathbf{L})\mathbf{v}_i^H = \mathbf{L}_{\boldsymbol{\theta}} \boldsymbol{e}_i^H. \tag{18b}$$

One may again verify $(\lambda_i^H \widehat{\mathbf{I}} - \widehat{\mathbf{L}})\widehat{\mathbf{v}}_i^H = \mathbf{0}$. On the other hand, when the harmonic forcing is resonant with some of the angular frequencies of $\mathbf{L}$ then, analogous to the case of parameter modes, one defines,

$$(\boldsymbol{\gamma}_i^H)_j = \begin{cases} \mathbf{w}_j^\dagger \mathbf{L}_{\boldsymbol{\theta}} \boldsymbol{e}_i^H, & \textit{if } \lambda_i^H = \lambda_j^C \\ 0, & \textit{otherwise}, \end{cases} \tag{19}$$

where, $(\boldsymbol{\gamma}_i^H)_j$ is the j^{th} component of the vector $\boldsymbol{\gamma}_i^H$. The extended modes are then given by

$$\widehat{\mathbf{v}}_i^H = \left[\mathbf{v}_i^H;\ \mathbf{0};\ \boldsymbol{e}_i^H\right], \quad \forall i \in 1, \ldots h. \tag{20a}$$
$$(\lambda_i^H \mathbf{I} - \mathbf{L})\mathbf{v}_i^H = \mathbf{L}_{\boldsymbol{\theta}} \boldsymbol{e}_i^H - \mathbf{V}\boldsymbol{\gamma}_i^H, \tag{20b}$$
$$\mathbf{w}_j^\dagger \mathbf{v}_i^H = 0, \qquad \textit{if } \lambda_i^H = \lambda_j^C, \tag{20c}$$

In the resonant case, this satisfies the generalized eigenvalue equation $(\lambda_i^H \widehat{\mathbf{I}} - \widehat{\mathbf{L}})^2 \widehat{\mathbf{v}}_i^H = \mathbf{0}$. Again, equation (20) reduces to equation (18) when the forcing is non-resonant. Henceforth equation (20) will be used as the definition of the forcing modes. The two sets of vectors $\boldsymbol{\gamma}_i^P$ and $\boldsymbol{\gamma}_i^H$ are represented together as matrices,

$$\boldsymbol{\Gamma}_P = [\boldsymbol{\gamma}_1^P, \boldsymbol{\gamma}_2^P, \ldots, \boldsymbol{\gamma}_l^P], \tag{21a}$$
$$\boldsymbol{\Gamma}_H = [\boldsymbol{\gamma}_1^H, \boldsymbol{\gamma}_2^H, \ldots, \boldsymbol{\gamma}_h^H], \tag{21b}$$

The critical eigenspace of $\widehat{\mathbf{L}}$ can be represented by three matrices

$$\widehat{\mathbf{V}}_C = \begin{bmatrix} \mathbf{V} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \quad \widehat{\mathbf{V}}_P = \begin{bmatrix} \mathbf{V}_P \\ \mathbf{I}_l \\ \mathbf{0} \end{bmatrix}, \quad \widehat{\mathbf{V}}_H = \begin{bmatrix} \mathbf{V}_H \\ \mathbf{0} \\ \mathbf{I}_h \end{bmatrix}, \tag{22}$$

where, $\mathbf{V}$ is the matrix of the critical eigenvectors of $\mathbf{L}$. The matrix $\mathbf{V}_P$ comprises of the l vectors $\mathbf{v}_i^P$ of the parameter modes, and, $\mathbf{V}_H$ comprises of the h vectors $\mathbf{v}_i^H$ of the forcing modes. The matrix of vectors of the extended eigenspace is denoted by, $\widehat{\mathbf{V}} = [\widehat{\mathbf{V}}_C, \widehat{\mathbf{V}}_P, \widehat{\mathbf{V}}_H]$ with the individual vectors denoted as $\widehat{\mathbf{v}}_i,\ i \in 1, 2, \ldots, m$.

2.2 Subspace structure for $\widehat{\mathbf{L}}^\dagger$

The eigenspace structure for the adjoint problem is important to consider since these vectors define the projection operators into the extended center subspace. The critical eigenmodes of the $\widehat{\mathbf{L}}^\dagger$ can be extended via the following definition,

$$\widehat{\mathbf{w}}_i^C = [\mathbf{w}_i; \boldsymbol{\zeta}_i^P; \boldsymbol{\zeta}_i^H], \tag{23a}$$
$$(\lambda_i^{C*}\mathbf{I} - \mathbf{L}^\dagger)\mathbf{w}_i = 0, \qquad \forall i \in 1, \dots n, \tag{23b}$$

with vectors $\boldsymbol{\zeta}_i^P$ and $\boldsymbol{\zeta}_i^H$ defined such that the j^{th} component of $\boldsymbol{\zeta}_i^P$ given by,

$$(\zeta_i^P)_j = \begin{cases} (\mathbf{L}_{\boldsymbol{\nu}}^\dagger \mathbf{w}_i)_j/(\lambda_i^{C*} - \lambda_j^{P*}); & \textit{if } \lambda_i^{C*} \neq \lambda_j^{P*}, \\ 0; & \textit{if } \lambda_i^{C*} = \lambda_j^{P*}, \end{cases} \tag{24}$$

$\forall j \in 1, \dots, l$, and, the j^{th} component of $\boldsymbol{\zeta}_i^H$ is given by,

$$(\zeta_i^H)_j = \begin{cases} (\mathbf{L}_{\boldsymbol{\theta}}^\dagger \mathbf{w}_i)_j/(\lambda_i^{C*} - \lambda_j^{H*}); & \textit{if } \lambda_i^{C*} \neq \lambda_j^{H*}, \\ 0; & \textit{if } \lambda_i^{C*} = \lambda_j^{H*}, \end{cases} \tag{25}$$

$\forall j \in 1, \dots, h$. Here, $(\mathbf{L}_{\boldsymbol{\nu}}^\dagger \mathbf{w}_i)_j$ refers to the j^{th} component of the vector $(\mathbf{L}_{\boldsymbol{\nu}}^\dagger \mathbf{w}_i)$, and likewise for $(\mathbf{L}_{\boldsymbol{\theta}}^\dagger \mathbf{w}_i)_j$. It may be verified that the definition is consistent with the eigenvalue equation, $(\lambda_i^{C*}\widehat{\mathbf{I}} - \widehat{\mathbf{L}}^\dagger)\widehat{\mathbf{w}}_i^C = 0$, for the non-resonant cases, and satisfies generalized eigenvalue equation, $(\lambda_i^{C*}\widehat{\mathbf{I}} - \widehat{\mathbf{L}}^\dagger)^2\widehat{\mathbf{w}}_i^C = 0$, for the resonant cases.

The adjoint parameter modes are given simply by,

$$\widehat{\mathbf{w}}_i^P = \left[\mathbf{0};\ \boldsymbol{e}_i^P;\ \mathbf{0}\right], \qquad \forall i \in 1, \dots l, \tag{26}$$

which satisfies $(\lambda_i^{P*}\widehat{\mathbf{I}} - \widehat{\mathbf{L}}^\dagger)\widehat{\mathbf{w}}_i^P = 0$. And the adjoint forcing modes are given by,

$$\widehat{\mathbf{w}}_i^H = \left[\mathbf{0};\ \mathbf{0};\ \boldsymbol{e}_i^H\right], \qquad \forall i \in 1, \dots h, \tag{27}$$

which satisfies $(\lambda_i^{H*}\widehat{\mathbf{I}} - \widehat{\mathbf{L}}^\dagger)\widehat{\mathbf{w}}_i^H = 0$.

The extended eigenspace of $\widehat{\mathbf{L}}^\dagger$ can be represented with matrices as

$$\widehat{\mathbf{W}}_C = \begin{bmatrix} \mathbf{W} \\ \mathbf{Z}_P \\ \mathbf{Z}_H \end{bmatrix}, \quad \widehat{\mathbf{W}}_P = \begin{bmatrix} \mathbf{0} \\ \mathbf{I}_l \\ \mathbf{0} \end{bmatrix}, \quad \widehat{\mathbf{W}}_H = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{I}_h \end{bmatrix}, \tag{28}$$

where, $\mathbf{W}$ is the matrix of the adjoint critical eigenvectors of $\mathbf{L}^\dagger$. The matrix $\mathbf{Z}_P$ comprises of the vectors $\boldsymbol{\zeta}_i^P$ and, $\mathbf{Z}_H$ comprises of the vectors $\boldsymbol{\zeta}_i^H$. The matrix of vectors of the extended (adjoint) center subspace is denoted by, $\widehat{\mathbf{W}} = [\widehat{\mathbf{W}}_C, \widehat{\mathbf{W}}_P, \widehat{\mathbf{W}}_H]$ with the individual vectors denoted as $\widehat{\mathbf{w}}_i$, $i \in 1, 2, \dots, m$. It can be verified that,

$$\widehat{\mathbf{W}}^\dagger \widehat{\mathbf{V}} = \mathbf{I}_m. \tag{29}$$

This is the analogue of equation (6), and is simply a statement of the biorthogonality relation between the direct and adjoint eigenvectors in the extended center subspace.

Finally, we have the generalized eigenvalue relations,

$$\widehat{\mathbf{L}}\widehat{\mathbf{V}} = \widehat{\mathbf{V}}\widehat{\mathbf{K}}, \qquad \widehat{\mathbf{L}}^\dagger\widehat{\mathbf{W}} = \widehat{\mathbf{W}}\widehat{\mathbf{K}}^\dagger \tag{30}$$

with,

$$\widehat{\mathbf{K}} = \widehat{\mathbf{W}}^\dagger\widehat{\mathbf{L}}\widehat{\mathbf{V}} = \begin{bmatrix} \boldsymbol{\Lambda}_C & \boldsymbol{\Gamma}_P & \boldsymbol{\Gamma}_H \\ \mathbf{0} & \boldsymbol{\Lambda}_P & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{\Lambda}_H \end{bmatrix}, \tag{31}$$

being the reduced matrix representing the action of $\widehat{\mathbf{L}}$ in the extended center subspace. Obviously $\widehat{\mathbf{K}} \in \mathbb{C}^{m \times m}$ and the diagonal elements of $\widehat{\mathbf{K}}$ represent the eigenvalues of the reduced system, that are the critical eigenvalues of the extended system.

Note that due to the special structure of the adjoint parameter and forcing modes, the projection of the extended nonlinearity $\widehat{\mathcal{N}'}$ onto the subspace spanned by the (direct) parameter and forcing modes vanishes identically. That is to say,

$$\widehat{\mathbf{W}}_P^\dagger\widehat{\mathcal{N}'} = \mathbf{0}, \qquad \widehat{\mathbf{W}}_H^\dagger\widehat{\mathcal{N}'} = \mathbf{0}. \tag{32}$$

This completes the extension of the system and the subspace structure for both the direct and adjoint systems can be readily identified. The extension of the tangent space is summarized in algorithms 1 and 2.

While the structure of the extended subspaces has been evaluated, reformulation of the problem as an extended system may have detrimental practical implications when considering large systems. Practical methods for solving such systems often rely on numerical algorithms that exploit inherent

Algorithm 1 Extended tangent space ($\widehat{\mathbf{L}}$)

Require: System: $\partial\mathbf{u}/\partial t = \mathbf{L}\mathbf{u} + \mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b)$.
Ensure: $\mathcal{N}(\mathbf{0}, \boldsymbol{\mu}^b) = \mathbf{0}$; $\partial\mathcal{N}/\partial\mathbf{u} = \mathbf{0}$.

Require: Tangent Space: $\mathbf{W}^\dagger\mathbf{L}\mathbf{V} = \boldsymbol{\Lambda}_C$.
  /* *Center subspace dimension: n.* */
  Construct: Linear operators $\mathbf{L}_{\boldsymbol{\nu}} \in \mathbb{C}^{N\times l}$ and $\mathbf{L}_{\boldsymbol{\theta}} \in \mathbb{C}^{N\times m}$ for extended variables $\boldsymbol{\nu}, \boldsymbol{\theta}$.

Require: Diagonal submatrices: $\boldsymbol{\Lambda}_P, \boldsymbol{\Lambda}_H$.
  /* *Build: $\boldsymbol{\Gamma}_P \in \mathbb{C}^{n\times l}$; $\boldsymbol{\Gamma}_H \in \mathbb{C}^{n\times h}$.* */
  for $(i \in 1, \dots, l)$ **do**
    for $(j \in 1, \dots, n)$ **do**
      if $(\lambda_i^P = \lambda_j^C)$ **then**
        $(\boldsymbol{\Gamma}_P)_{j,i} = \mathbf{w}_j^\dagger \mathbf{L}_{\boldsymbol{\nu}} \boldsymbol{e}_i^P$,
      else
        $(\boldsymbol{\Gamma}_P)_{j,i} = 0$,
      end if
    end for
  end for

  /* *Solve $\mathbf{V}_P$* */
  $\mathbf{L}\mathbf{V}_P + \mathbf{L}_{\boldsymbol{\nu}}\mathbf{I}_l = \mathbf{V}_P\boldsymbol{\Lambda}_P + \mathbf{V}\boldsymbol{\Gamma}_P$, /* *Eq. (17)*
  /* *with singular components removed.* */
  for $(i \in 1, \dots, h)$ **do**
    for $(j \in 1, \dots, n)$ **do**
      if $(\lambda_i^H = \lambda_j^C)$ **then**
        $(\boldsymbol{\Gamma}_H)_{j,i} = (\boldsymbol{\gamma}_i^H)_j = \mathbf{w}_j^\dagger \mathbf{L}_{\boldsymbol{\theta}} \boldsymbol{e}_i^H$,
      else
        $(\boldsymbol{\Gamma}_H)_{j,i} = (\boldsymbol{\gamma}_i^H)_j = 0$.
      end if
    end for
  end for

  /* *Solve $\mathbf{V}_H$* */
  $\mathbf{L}\mathbf{V}_H + \mathbf{L}_{\boldsymbol{\theta}}\mathbf{I}_h = \mathbf{V}_H\boldsymbol{\Lambda}_H + \mathbf{V}\boldsymbol{\Gamma}_H$, /* *Eq. (20)*
  /* *with singular components removed.* */
  Construct: $\widehat{\mathbf{V}}$ /* *Eq. (22)*
  Construct: $\widehat{\mathbf{K}}$ /* *Eq. (31)*
  return $\widehat{\mathbf{K}}, \widehat{\mathbf{V}}$.

structure of the problem (Hermicity for conjugate-gradient solvers for example). In such a scenario, additional degrees of freedom can potential destroy the system symmetries and may require substantial changes in the numerical approach to the system solution. This would be particularly true of asymptotic approaches that require resolvent calculations [1, 2, 3, 4, 5, 6, 7, 8, 9], which itself is a computationally challenging task for

Algorithm 2 Extended tangent space ($\widehat{\mathbf{L}}^\dagger$)

Require: System: $\partial\mathbf{u}/\partial t = \mathbf{L}\mathbf{u} + \mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b)$.
Ensure: $\mathcal{N}(\mathbf{0}, \boldsymbol{\mu}^b) = \mathbf{0}$; $\partial\mathcal{N}/\partial\mathbf{u} = \mathbf{0}$.

Require: Tangent Space: $\mathbf{W}^\dagger\mathbf{L}\mathbf{V} = \boldsymbol{\Lambda}_C$.
  /* *Center subspace dimension: n.* */
  Construct: Linear operators $\mathbf{L}_{\boldsymbol{\nu}} \in \mathbb{C}^{N\times l}$ and $\mathbf{L}_{\boldsymbol{\theta}} \in \mathbb{C}^{N\times m}$ for extended variables $\boldsymbol{\nu}, \boldsymbol{\theta}$.

Require: Diagonal submatrices: $\boldsymbol{\Lambda}_P, \boldsymbol{\Lambda}_H$.

  /* *Build: $\mathbf{Z}_P \in \mathbb{C}^{l\times n}$; $\mathbf{Z}_H \in \mathbb{C}^{h\times n}$.* */
  for $(i \in 1, \dots, n)$ **do**
    for $(j \in 1, \dots, l)$ **do**
      if $(\lambda_j^P = \lambda_i^C)$ **then**
        $(\mathbf{Z}_P)_{j,i} = 0$,
      else
        $(\mathbf{Z}_P)_{j,i} = (\mathbf{L}_{\boldsymbol{\nu}}^\dagger \mathbf{w}_i)_j / (\lambda_i^{C*} - \lambda_j^{P*})$,
      end if
    end for
    for $(j \in 1, \dots, h)$ **do**
      if $(\lambda_j^H = \lambda_i^C)$ **then**
        $(\mathbf{Z}_H)_{j,i} = 0$,
      else
        $(\mathbf{Z}_H)_{j,i} = (\mathbf{L}_{\boldsymbol{\theta}}^\dagger \mathbf{w}_i)_j / (\lambda_i^{C*} - \lambda_j^{H*})$,
      end if
    end for
  end for
  Construct: $\widehat{\mathbf{W}}$ /* *Eq. (28)*
  return $\widehat{\mathbf{W}}$.

large systems. Such a scenario would be highly undesireable. However, as it turns out, at least some of the parameterizations involved in the evaluation of center-manifolds do not in fact require the use of the extended system after the initial evaluation of the extended tangent space. All higher order calculations can be performed using the original linear system $\mathbf{L}$. This is clarified in the next section where the parameterization and its asymptotic terms are explicitly built.

3 Center Manifold for the Extended System

In the context of invariant manifold theory, several previous approaches for the approximation of such manifolds can be effectively viewed as instances of what is referred to as the parametrization method. The framework was introduced in a series of papers by Cabré et al. [10, 11, 2], although, a particular instance of such an approach was presented earlier by Coullet and Spiegel [1] for the evaluation of the normal form of center manifolds. Nevertheless, subsequent works by Carini, Auteri, and Giannetti [5], Jain and Haller [8], Negi [9] can be fruitfully viewed through the lens of the parameterization method. Within this framework the reduced representation of the invariant manifold being approximated can be chosen in an arbitrary manner depending on the particular requirements of the reduction being sought. Amongst the plethora of arbitrary choices, the normal form is an important representation that is often sought since it results in the simplest form of the reduced equations. In the next subsection the normal form representation for the center-manifold of the extended system is deduced.

To simplify notation for multiple indices used hereafter, a multi-dimensional index set $\boldsymbol{\mathcal{I}}\{(k),[a,b]\}$ is introduced, which is an *ordered* set of all k-tuples $(i_1, i_2 \ldots, i_k)$, defined as,

$$\begin{aligned}\boldsymbol{\mathcal{I}}\{(k),[a,b]\} :=\big\{&(i_1, i_2 \ldots, i_k) \mid a, b \in \mathbb{N},\\ &i_1, i_2 \ldots, i_k \in \mathbb{N},\\ &a \leq i_i \leq b,\ i_1 \leq i_2 \leq b,\\ &\ldots,\ i_{k-1} \leq i_k \leq b.\big\},\end{aligned} \tag{33}$$

where, $i_1, i_2, \ldots, i_k$, are the index variables and $[a, b]$ specify the lower and upper bounds of the indices. Obviously the indices i_1 *etc.* are dummy indices however their relative position is important since the higher index always starts from the previous index. The set is ordered in the sense that the first element of the $\boldsymbol{\mathcal{I}}\{(k),[a,b]\}$ is always $(i_1, i_2 \ldots, i_k) = (a, a, \ldots, a)$, the last element is $(i_1, i_2 \ldots, i_k) = (b, b, \ldots, b)$ and it is the last index (i_k) which iterates the fastest. Hence the index set $\boldsymbol{\mathcal{I}}\{(3),[1,3]\}$ is the set containing the tuples,

$$\begin{aligned}(i_1, i_2, i_3) \in \{&(1,1,1), (1,1,2), (1,1,3),\\ &(1,2,2), (1,2,3), (1,3,3),\\ &(2,2,2), (2,2,3), (2,3,3),\\ &(3,3,3)\} ,\end{aligned}$$

in that order. If an element $(i_1, i_2, \ldots, i_k)$ precedes an element $(j_1, j_2, \ldots, j_k)$ in ordering it is denoted as $(i_1, i_2, \ldots, i_k) \prec (j_1, j_2, \ldots, j_k)$ while, if $(i_1, i_2, \ldots, i_k)$ succeeds $(j_1, j_2, \ldots, j_k)$ in the ordering it is denoted as $(i_1, i_2, \ldots, i_k) \succ (j_1, j_2, \ldots, j_k)$.

In a similar vein, multiple nested summations are denoted as

$$\sum_{(k)}^{[a,b]} = \sum_{i_1=a}^{b} \sum_{i_2=i_1}^{b} \cdots \sum_{i_k=i_{k-1}}^{b} \tag{34}$$

As before, the higher index starts from the previous index. In principle the ordering of the tuples in a particular index set does not matter to the overall results however, keeping the ordering in mind helps to elucidate the structure of the asymptotic problem as will be shown subsequently.

For the asymptotic expansions we will typically have $a = 1$ and $b = m$, where m is the dimension of the manifold. For such cases the number of k-tuples in $\boldsymbol{\mathcal{I}}\{(k),[1,m]\}$ is analytically known to be $T_m^k = C(m + k - 1, k)$, where,

$$C(p,q) = \binom{p}{q} = \frac{p!}{q!(p-q)!}, \tag{35}$$

defines the binomial coefficient of p and q.

3.1 Normal Form

One may find the exposition on the implementation of the parameterization method in previous works of Haro et al. [7] and Jain and Haller [8]. To be clear, the application of the parameterization method to an extended system like equation (10) follows the same steps as outlined in those earlier works. However, many of the essential steps are repeated here to bring out the structure of system of equations at each asymptotic order in the context of system extension.

Consider the extended system as defined by equation (10), represented in a compact form by equation (12), which has an m-dimensional critical subspace ($m = n + l + h$). Assume $\mathbf{z} = [z_1; z_2; \ldots; z_m]$ is an m-dimensional vector of coordinates that parameterize the center-manifold of the extended system. The evolution of the system along the center-manifold in the parameterized coordinates is assumed to be given by

$$\begin{aligned}\dot{\mathbf{z}} &= \mathcal{G}(\mathbf{z}),\\ \mathcal{G} &: \mathbb{C}^m \to \mathbb{C}^m.\end{aligned} \tag{36}$$

$\mathcal{G}$ will also be referred to as the reduced representation (of the extended system). The system solution on the center-manifold corresponding to a point $\mathbf{z}$ of the parameterized system is given by $\mathcal{Y}(\mathbf{z})$, which is map from the parameterized coordinates to the full system state,

$$\mathcal{Y} : \mathbb{C}^m \rightarrow \mathbb{C}^M. \tag{37}$$

At this stage neither $\mathcal{G}(\mathbf{z})$ nor $\mathcal{Y}(\mathbf{z})$ have been specified in any form, since, the parameterization can be arbitrarily chosen. In order to obtain a normal form of the reduced system one may expand both $\mathcal{G}(\mathbf{z})$ and $\mathcal{Y}(\mathbf{z})$ as a formal power series in $\mathbf{z}$ and constrain the asymptotic form of $\mathcal{G}$ to be in a normal form for the parameterized coordinates $\mathbf{z}$. The asymptotic form of $\mathcal{Y}$ is obtained by introducing the asymptotic expansions into equation (12) and solving the terms order-by-order.

In what follows, Einstein summation is *not* assumed for repeated indices unless otherwise specified. We assume the power series expansions as,

$$\begin{aligned}\mathcal{G}(\mathbf{z}) = \sum_{(1)}^{[1,m]} \mathbf{g}_{i_1} z_{i_1} + \sum_{(2)}^{[1,m]} \mathbf{g}_{i_1,i_2} z_{i_1} z_{i_2} + \\ \sum_{(3)}^{[1,m]} \mathbf{g}_{i_1,i_2,i_3} z_{i_1} z_{i_2} z_{i_3} + \dots,\end{aligned} \tag{38}$$

and,

$$\begin{aligned}\mathcal{Y}(\mathbf{z}) = \sum_{(1)}^{[1,m]} \widehat{\mathbf{y}}_{i_1} z_{i_1} + \sum_{(2)}^{[1,m]} \widehat{\mathbf{y}}_{i_1,i_2} z_{i_1} z_{i_2} + \\ \sum_{(3)}^{[1,m]} \widehat{\mathbf{y}}_{i_1,i_2,i_3} z_{i_1} z_{i_2} z_{i_3} + \dots.\end{aligned} \tag{39}$$

Here the terms $\mathbf{g}_{i_1,i_2,i_3,\dots} \in \mathbb{C}^m$, while, $\widehat{\mathbf{y}}_{i_1,i_2,i_3,\dots} \in \mathbb{C}^M$. Given the asymptotic form of $\mathcal{Y}(\mathbf{z})$, one automatically obtains the nonlinear term $\widehat{\mathcal{N}'}$ as a power series in $\mathbf{z}$. This can be represented as,

$$\begin{aligned}\widehat{\mathcal{N}'} = \sum_{(2)}^{[1,m]} \widehat{\mathbf{h}}_{i_1,i_2} z_{i_1} z_{i_2} + \\ \sum_{(3)}^{[1,m]} \widehat{\mathbf{h}}_{i_1,i_2,i_3} z_{i_1} z_{i_2} z_{i_3} + \dots,\end{aligned} \tag{40}$$

which is obtained via the collection of terms with similar powers.

Substitution of the power series expansions into equation (12) results in

$$\begin{aligned}\sum_{\alpha=1}^{m} \frac{\partial}{\partial z_\alpha} \Big(\sum_{(1)}^{[1,m]} \widehat{\mathbf{y}}_{i_1} z_{i_1} + \dots \Big) \Big(\sum_{(1)}^{[1,m]} g^{\alpha}_{j_1} z_{j_1} + \dots \Big) = \\ \widehat{\mathbf{L}} \Big(\sum_{(1)}^{[1,m]} \widehat{\mathbf{y}}_{i_1} z_{i_1} + \dots \Big) + \Big(\sum_{(2)}^{[1,m]} \widehat{\mathbf{h}}_{i_1,i_2} z_{i_1} z_{i_2} + \dots \Big).\end{aligned} \tag{41}$$

Here $g^{\alpha}_{j_1}$ represents the α^{th} element of the vector $\mathbf{g}_{j_1}$. At order k, one obtains $m \times T^k_m$ unknowns which define the elements of the vectors $\mathbf{g}_{i_1,\dots,i_k}$, and the corresponding T^k_m unknown vectors $\widehat{\mathbf{y}}_{i_1,\dots,i_k}$. The definitions of the two sets of unknowns are interdependent since the form of the reduced representation $\mathcal{G}(\mathbf{z})$ determines the map $\mathcal{Y}(\mathbf{z})$ from the parameterized coordinates to the full solution (and vice versa).

3.1.1 Asymptotic terms of $\mathcal{O}(|x|)$

At first order the non-linear terms do not contribute and one simply obtains,

$$\sum_{(1)}^{[1,m]} \Big(\sum_{\alpha=1}^{m} g^{\alpha}_{i_1} \widehat{\mathbf{y}}_\alpha - \widehat{\mathbf{L}} \widehat{\mathbf{y}}_{i_1} \Big) z_{i_1} = \mathbf{0}. \tag{42}$$

Here, the first restrictions on the asymptotic form of $\mathcal{G}$ are applied, associating the unknown terms $\widehat{\mathbf{y}}_{i_1}$, $\mathbf{g}_{i_1}$ *etc.* with the generalized eigenvalue problem such that,

$$\big[\widehat{\mathbf{y}}_{i_1}, \widehat{\mathbf{y}}_{i_2}, \dots, \widehat{\mathbf{y}}_{i_m}\big] = \widehat{\mathbf{V}}, \tag{43a}$$

$$\big[\mathbf{g}_{i_1}, \mathbf{g}_{i_2}, \dots, \mathbf{g}_{i_m}\big] = \widehat{\mathbf{K}}. \tag{43b}$$

This reduces equation (42) to equation (30) which, then satisfies the tangency condition of the invariant manifold, *i.e.* , at the origin the center-manifold is tangent to the center subspace. Subsequently, $\widehat{K}_{i,j}$ will be used for all the first order terms of the type $\mathbf{g}^i_j$ and the first order vectors $\widehat{\mathbf{y}}_i$ will be replaced by the respective eigenvectors $\widehat{\mathbf{v}}_i$. The reduced matrix $\widehat{\mathbf{K}}$ is responsible for the cross-coupling of different $\widehat{\mathbf{y}}_{i_1,\ldots,i_k}$ at each order of the asymptotic expansion.

3.1.2 Asymptotic terms of $\mathcal{O}(|\mathbf{z}|^2)$

At second order the non-linear terms contribute and one obtains the following equation,

$$\begin{aligned}&\sum_{(2)}^{[1,m]}\sum_{j=1}^{m}\Big(\widehat{K}_{i_1,j}z_{i_2}z_j+\widehat{K}_{i_2,j}z_{i_1}z_j\Big)\widehat{\mathbf{y}}_{i_1,i_2}+\\&\sum_{(2)}^{[1,m]}\sum_{j=1}^{m}g^j_{i_1,i_2}\widehat{\mathbf{v}}_j z_{i_1}z_{i_2}-\sum_{(2)}^{[1,m]}\widehat{\mathbf{L}}\widehat{\mathbf{y}}_{i_1,i_2}z_{i_1}z_{i_2}\\&=\sum_{(2)}^{[1,m]}\widehat{\mathbf{h}}_{i_1,i_2}z_{i_1}z_{i_2}.\end{aligned}\tag{44}$$

Collecting the expressions for a particular asymptotic term $z_a z_b$, where $(a,b)\in\boldsymbol{\mathcal{I}}\{(2),[1,m]\}$, equation (44) reduces to,

$$\begin{aligned}&\Big[\sum_{i=1}^{a}\widehat{K}_{i,b}\widehat{\mathbf{y}}_{i,a}+\sum_{i=1}^{b}\widehat{K}_{i,a}\widehat{\mathbf{y}}_{i,b}-\sum_{i=1}^{a}\widehat{K}_{i,b}\widehat{\mathbf{y}}_{i,a}\delta^a_b\Big]+\\&\Big[\sum_{i=a}^{m}\widehat{K}_{i,b}\widehat{\mathbf{y}}_{a,i}+\sum_{i=b}^{m}\widehat{K}_{i,a}\widehat{\mathbf{y}}_{b,i}-\sum_{i=b}^{m}\widehat{K}_{i,a}\widehat{\mathbf{y}}_{b,i}\delta^a_b\Big]\\&+\sum_{j=1}^{m}g^j_{a,b}\widehat{\mathbf{v}}_j-\widehat{\mathbf{L}}\widehat{\mathbf{y}}_{a,b}=\widehat{\mathbf{h}}_{a,b}.\end{aligned}\tag{45}$$

At second order one obtains T^2_m equations, one for each tuple $(a,b)\in\boldsymbol{\mathcal{I}}\{(2),[1,m]\}$, which collectively form the system of equations for all $\widehat{\mathbf{y}}_{i_1,i_2}$ vectors that needs to be solved. Here, several properties of the structure of equation (45) need to be pointed out that have important practical implications.

First, the various unknown vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ are coupled due to the off-diagonal terms in $\widehat{\mathbf{K}}$, which appear in the first two bracketted terms in equation (45). Recall that we are considering the tuples in our index set to be ordered so that the vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ can also be ordered in the same manner. For any index $(a,b)\in\boldsymbol{\mathcal{I}}\{(2),[1,m]\}$ (and assuming $b>a$) the bracketted terms in equation (45) can be rewritten as,

$$\begin{aligned}&\overbrace{\sum_{i=1}^{a}\Big(\widehat{K}_{i,b}\widehat{\mathbf{y}}_{i,a}-\widehat{K}_{i,b}\widehat{\mathbf{y}}_{i,a}\delta^a_b\Big)}^{P1}+\overbrace{\sum_{i=1}^{a-1}\widehat{K}_{i,a}\widehat{\mathbf{y}}_{i,b}}^{P2}+\\&\overbrace{\widehat{K}_{a,a}\widehat{\mathbf{y}}_{a,b}}^{D1}+\overbrace{\sum_{i=a+1}^{b}\widehat{K}_{i,a}\widehat{\mathbf{y}}_{i,b}}^{S1}+\\&\overbrace{\sum_{i=a}^{b-1}\widehat{K}_{i,b}\widehat{\mathbf{y}}_{a,i}}^{P3}+\overbrace{\widehat{K}_{b,b}\widehat{\mathbf{y}}_{a,b}}^{D2}+\overbrace{\sum_{i=b+1}^{m}\widehat{K}_{i,b}\widehat{\mathbf{y}}_{a,i}}^{S2}+\\&\overbrace{\sum_{i=b}^{m}\Big(\widehat{K}_{i,a}\widehat{\mathbf{y}}_{b,i}-\widehat{K}_{i,a}\widehat{\mathbf{y}}_{b,i}\delta^a_b\Big)}^{S3}.\end{aligned}\tag{46}$$

The terms marked as $P1,P2,P3$ are terms where elements of $\widehat{\mathbf{K}}$ multiply vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ such that $(i_1,i_2)\prec(a,b)$. The terms marked $D1,D2$ are those where elements of $\widehat{\mathbf{K}}$ multiply only with $\widehat{\mathbf{y}}_{a,b}$, and the terms marked as $S1,S2,S3$ are those where $\widehat{\mathbf{K}}$ elements multiply vectors succeeding $\widehat{\mathbf{y}}_{a,b}$, *i.e.* , $(i_1,i_2)\succ(a,b)$. Notice that all the succeeding terms in $S1,S2,S3$ are always multiplied by elements of the $\widehat{\mathbf{K}}$ matrix that are below the main diagonal. However, since $\widehat{\mathbf{K}}$ has an upper triangular structure, all terms marked $S1,S2,S3$ vanish. Therefore the equation resulting from any polynomial term $z_a z_b$ couples the vector $\widehat{\mathbf{y}}_{a,b}$ only with those that preceed it in the ordering, *i.e.* , with $\widehat{\mathbf{y}}_{i_1,i_2}\prec\widehat{\mathbf{y}}_{a,b}$. Thus the resulting system of equations for the T^2_m unknown vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ at second order is lower triangular and even though the fields are coupled with each other, they can still be solved for sequentially one at a time with back substitution. The splitting of terms in equation (46) is done for $b>a$ however, the same conclusion applies even for the case of $b=a$ and the resulting system of equations is always lower triangular.

A visual representation of this cross coupling is shown in figure 1. For a given index $(a,b)\in\boldsymbol{\mathcal{I}}\{(2),[1,m]\}$, the elements of the matrix

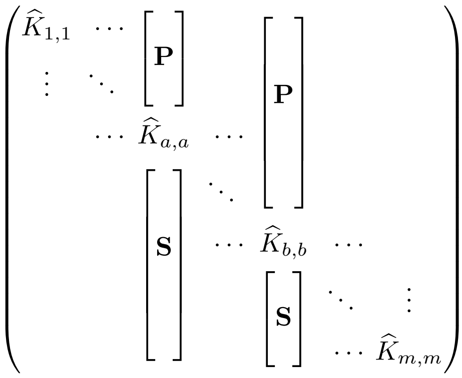

Fig. 1: Coupling of the vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ with the $\widehat{\mathbf{K}}$ matrix for a given (a, b).

$\widehat{\mathbf{K}}$ marked by $\mathbf{P}$ multiply vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ that precede $\widehat{\mathbf{y}}_{a,b}$ in the ordering. Similarly, the elements marked by $\mathbf{S}$ multiply the vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ which succeed $\widehat{\mathbf{y}}_{a,b}$ in the ordering. The diagonal terms $\widehat{K}_{a,a}$ and $\widehat{K}_{b,b}$ always multiply $\widehat{\mathbf{y}}_{a,b}$. Obviously, when $\widehat{\mathbf{K}}$ is upper triangular, the contributions of the succeeding elements will always vanish. This lower triangular structure of the resulting system of equations for $\widehat{\mathbf{y}}_{i_1,i_2}$ is not confined to the second-order approximation but in fact persists even for higher order asymptotic approximations so that at all orders, the system of equations can always be solved sequentially, one vector at a time. This is an important aspect particularly for large systems since the computational cost of a simultaneous coupled system solve can be prohibitive (as T_m^k becomes large).

To keep the notation compact, the non-vanishing off-diagonal coupling terms will be represented as $\widehat{\mathbf{h}}^O_{a,b}$, defined for second order as,

$$\widehat{\mathbf{h}}^O_{a,b} = \begin{cases} \sum_{i=1}^{a} \left(\widehat{K}_{i,b} \widehat{\mathbf{y}}_{i,a} - \widehat{K}_{i,b} \widehat{\mathbf{y}}_{i,a} \delta^a_b \right) \\ + \sum_{i=1}^{a-1} \widehat{K}_{i,a} \widehat{\mathbf{y}}_{i,b} + \sum_{i=a}^{b-1} \widehat{K}_{i,b} \widehat{\mathbf{y}}_{a,i}. \end{cases} \quad (47)$$

The resulting equation for $\widehat{\mathbf{y}}_{a,b}$ may be written as

$$[\widehat{\mathcal{R}}(\omega_{a,b})]^{-1} \widehat{\mathbf{y}}_{a,b} + \sum_{i=1}^{m} g^i_{a,b} \widehat{\mathbf{v}}_i = \widehat{\mathbf{h}}_{a,b} - \widehat{\mathbf{h}}^O_{a,b}, \quad (48a)$$

$$\omega_{a,b} = \widehat{K}_{a,a} + \widehat{K}_{b,b}, \quad (48b)$$

where, $\widehat{\mathcal{R}}(\omega)$ is the resolvent operator for the extended linear system defined as,

$$\widehat{\mathcal{R}}(\omega) = (\omega \widehat{\mathbf{I}} - \widehat{\mathbf{L}})^{-1}. \quad (49)$$

The resolvent operator is the key mathematical object defining the asymptotic expressions. $\widehat{\mathcal{R}}(\omega_{a,b})$ is singular when $\omega_{a,b}$ is an eigenvalue of $\widehat{\mathbf{L}}$, which is often referred to as a resonance condition. The normal form of the invariant manifold precisely resolves this singularity by requiring $\widehat{\mathbf{y}}_{a,b}$ to have a vanishing component along the singular direction. However, equation (45) still needs to be balanced along the singular directions, which can only be done by $\sum_{j=1}^{m} g^j_{a,b} \widehat{\mathbf{v}}_j$. Multiplying equation (48) with $\widehat{\mathbf{w}}^\dagger_j$ and requiring that $\widehat{\mathbf{w}}^\dagger_j \widehat{\mathbf{y}}_{a,b} = 0$ (in case of resonance), then provides us with the definition of $\mathbf{g}_{a,b}$,

$$g^j_{a,b} = \begin{cases} \widehat{\mathbf{w}}^\dagger_j (\widehat{\mathbf{h}}_{a,b} - \widehat{\mathbf{h}}^O_{a,b}); & \textit{if } \omega_{a,b} = \widehat{K}_{j,j}, \\ 0, & \textit{otherwise}, \end{cases} \quad (50)$$

where, $\widehat{\mathbf{w}}_j$ is the j^{th} adjoint eigenvector of $\widehat{\mathbf{L}}$ as defined earlier. Since the system is lower triangular the terms $\widehat{\mathbf{h}}^O_{a,b}$ are always known apriori when solving the system sequentially.

3.1.3 Asymptotic terms of $\mathcal{O}(|\mathbf{z}|^3)$

At third (and higher) order one also obtains contributions from the lower order terms. The sum of all contributions at third order can be written as,

$$\sum_{(3)}^{[1,m]} \sum_{j=1}^{m} \Big[\widehat{K}_{i_1,j} z_{i_2} z_{i_3} z_j + \widehat{K}_{i_2,j} z_{i_1} z_{i_3} z_j + \widehat{K}_{i_3,j} z_{i_1} z_{i_2} z_j \Big] \widehat{\mathbf{y}}_{i_1,i_2,i_3} + \sum_{(2)}^{[1,m]} \sum_{(2)}^{[1,m]} \left[g^{i_1}_{j_1,j_2} z_{i_2} z_{j_1} z_{j_2} + g^{i_2}_{j_1,j_2} z_{i_1} z_{j_1} z_{j_2} \right] \widehat{\mathbf{y}}_{i_1,i_2} + \sum_{j=1}^{m} \sum_{(3)}^{[1,m]} g^j_{i_1,i_2,i_3} \widehat{\mathbf{v}}_j z_{i_1} z_{i_2} z_{i_3} - \sum_{(3)}^{[1,m]} \widehat{\mathbf{L}} \widehat{\mathbf{y}}_{i_1,i_2,i_3} z_{i_1} z_{i_2} z_{i_3} = \sum_{(3)}^{[1,m]} \widehat{\mathbf{h}}_{i_1,i_2,i_3} z_{i_1} z_{i_2} z_{i_3}. \quad (51)$$

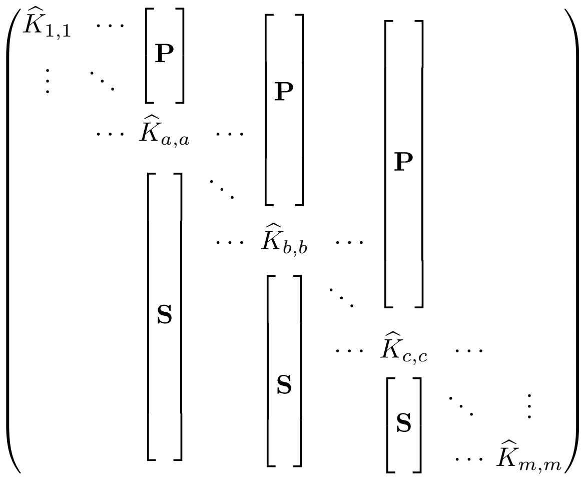

Fig. 2: Coupling of the vectors $\widehat{\mathbf{y}}_{i_1,i_2,i_3}$ with the $\widehat{\mathbf{K}}$ matrix for a given (a, b, c).

The $\widehat{\mathbf{K}}$ matrix is responsible for the cross contributions of different vectors $\widehat{\mathbf{y}}_{i_1,i_2,i_3}$ as seen from the first set of bracketted terms in equation (51). Again, for a given tuple $(a, b, c) \in \boldsymbol{\mathcal{I}}\{(3), [1, m]\}$, the coupling of $\widehat{\mathbf{K}}$ elements with vectors preceeding and succeeding $\widehat{\mathbf{y}}_{a,b,c}$ is depicted in figure 2. With $\widehat{\mathbf{K}}$ being upper triangular, the contribution of succeeding elements, $(i_1, i_2, i_3) \succ (a, b, c)$, vanishes and the resulting system of equations for $\widehat{\mathbf{y}}_{i_1,i_2,i_3}$ at third order is again lower triangular.

The second set of bracketted terms in equation (51) are contributions from the lower order asymptotic terms. For a given index (a, b, c), we represent the low order contributions as $\widehat{\mathbf{h}}^L_{a,b,c}$ and the contributions of the off-diagonal terms as $\widehat{\mathbf{h}}^O_{a,b,c}$ resulting in the compact expression for $\widehat{\mathbf{y}}_{a,b,c}$ as

$$[\widehat{\mathcal{R}}(\omega_{a,b,c})]^{-1}\widehat{\mathbf{y}}_{a,b,c} + \sum_{j=1}^{m} g^j_{a,b,c}\widehat{\mathbf{v}}_j = \widehat{\mathbf{h}}_{a,b,c} - \widehat{\mathbf{h}}^O_{a,b,c} - \widehat{\mathbf{h}}^L_{a,b,c}, \tag{52a}$$

$$\omega_{a,b,c} = \widehat{K}_{a,a} + \widehat{K}_{b,b} + \widehat{K}_{c,c} \tag{52b}$$

The resolution of the system singularities due to the resonance condition, $\omega_{a,b,c} = \widehat{K}_{j,j}$, provides us with the third order terms of the reduced representation,

$$g^j_{a,b,c} = \begin{cases} \widehat{\mathbf{w}}^\dagger_j(\widehat{\mathbf{h}}_{a,b,c} - \widehat{\mathbf{h}}^L_{a,b,c} - \widehat{\mathbf{h}}^O_{a,b,c}); & \textit{if, } \omega_{a,b,c} = \widehat{K}_{j,j} \\ 0, & \textit{otherwise,} \end{cases} \tag{53}$$

and $\widehat{\mathbf{w}}^\dagger_j\widehat{\mathbf{y}}_{a,b,c} = 0$ if the resonance condition holds.

3.1.4 Asymptotic terms of $\mathcal{O}(|\mathbf{z}|^{\boldsymbol{k}})$

For all subsequent orders the structure of the problem remains essentially the same as the third order terms. At any order k, one obtains a systems of equations for T^k_m vectors $\widehat{\mathbf{y}}_{i_1,\ldots,i_k}$ which is lower triangular and hence can be solved sequentially. The cross contributions of the vectors due to $\widehat{\mathbf{K}}$ has the same structure as in figure 2 but with k columns (possibly with repetitions) of $\widehat{\mathbf{K}}$ contributing. The lower order terms $\widehat{\mathbf{h}}^L_{i_1,\ldots,i_k}$ are always known a-priori. When solving for the unknown vectors sequentially the off-diagonal contributions $\widehat{\mathbf{h}}^O_{i_1,\ldots,i_k}$ are also known a-priori. For each index $(i_1, \ldots, i_k) \in \boldsymbol{\mathcal{I}}\{(k), [1, m]\}$ one obtains an equation similar to equation (52),

$$[\widehat{\mathcal{R}}(\omega_{i_1,\ldots,i_k})]^{-1}\widehat{\mathbf{y}}_{i_1,\ldots,i_k} + \sum_{j=1}^{m} g^j_{i_1,\ldots,i_k}\widehat{\mathbf{v}}_j = \widehat{\mathbf{h}}_{i_1,\ldots,i_k} - \widehat{\mathbf{h}}^O_{i_1,\ldots,i_k} - \widehat{\mathbf{h}}^L_{i_1,\ldots,i_k}, \tag{54a}$$

$$\omega_{i_1,\ldots,i_k} = \widehat{K}_{i_1,i_1} + \widehat{K}_{i_2,i_2} + \ldots + \widehat{K}_{i_k,i_k}, \tag{54b}$$

with the resonance condition arising whenever,

$$\omega_{i_1,\ldots,i_k} = \widehat{K}_{j,j}, \qquad j \in 1, \ldots, m, \tag{55}$$

which then defines the normal form coefficients,

$$g^j_{i_1,\ldots,i_k} = \begin{cases} \widehat{\mathbf{w}}^\dagger_j(\widehat{\mathbf{h}}_{i_1,\ldots,i_k} - \widehat{\mathbf{h}}^L_{i_1,\ldots,i_k} - \widehat{\mathbf{h}}^O_{i_1,\ldots,i_k}); & \textit{if, } \omega_{i_1,\ldots,i_k} = \widehat{K}_{j,j} \\ 0 = & \textit{otherwise,} \end{cases} \tag{56}$$

and, $\widehat{\mathbf{w}}^\dagger_j\widehat{\mathbf{y}}_{i_1,\ldots,i_k} = 0$ in case of resonance.

3.1.5 From the Extended System to the Standard System

For all orders $k \geq 2$, one needs to evaluate a resolvent equation of the type given by equation (54) for a given right hand side. The $\widehat{\cdot}$ symbol on all terms indicates that the resolvent needs to be evaluated for the extended system. This can potentially be a major drawback since numerical methods for large systems often rely on exploiting the inherent symmetry of the operators which may be destroyed by system extension. However, as it turns out, even though the resolvents are technically solved for the extended linear system $\widehat{\mathbf{L}}$, in fact only the resolvent solutions of the original linear operator $\mathbf{L}$ are required. This can be shown by examining the terms sequentially.

First we express equation (54) in its three component form for a generic vector $\widehat{\mathbf{y}} = [\mathbf{y}^{(1)}; \mathbf{y}^{(2)}; \mathbf{y}^{(3)}]$, for a particular frequency ω and generic right hand side, $\widehat{\mathbf{r}} = [\mathbf{r}^{(1)}; \mathbf{r}^{(2)}; \mathbf{r}^{(3)}]$ as,

$$(\omega\mathbf{I} - \mathbf{L})\mathbf{y}^{(1)} + \mathbf{L}_{\boldsymbol{\nu}}\mathbf{y}^{(2)} + \mathbf{L}_{\boldsymbol{\theta}}\mathbf{y}^{(3)} = \mathbf{r}^{(1)}, \tag{57a}$$

$$(\omega\mathbf{I} - \boldsymbol{\Lambda}_P)\mathbf{y}^{(2)} = \mathbf{r}^{(2)}, \tag{57b}$$

$$(\omega\mathbf{I} - \boldsymbol{\Lambda}_H)\mathbf{y}^{(3)} = \mathbf{r}^{(3)}. \tag{57c}$$

Clearly, $\mathbf{r}^{(2)} = \mathbf{0}$ and an ω that is non-resonant with $\boldsymbol{\lambda}^P$ implies $\mathbf{y}^{(2)} = \mathbf{0}$. Similarly, $\mathbf{r}^{(3)} = \mathbf{0}$ implies $\mathbf{y}^{(3)} = \mathbf{0}$ when ω is non-resonant with the external forcing frequencies $\boldsymbol{\lambda}^H$. In either case, if ω is resonant with the frequencies in $\boldsymbol{\lambda}^P$ or $\boldsymbol{\lambda}^H$, the condition for the removal of singularities, $\widehat{\mathbf{w}}_j^\dagger\widehat{\mathbf{y}} = \mathbf{0}$, again ensures $\mathbf{y}^{(2)} = \mathbf{0}$ and $\mathbf{y}^{(3)} = \mathbf{0}$. Therefore equation (57) simply reduces to,

$$[\mathcal{R}(\omega)]^{-1}\mathbf{y}^{(1)} = \mathbf{r}^{(1)}; \quad \mathcal{R}(\omega) = (\omega\mathbf{I} - \mathbf{L})^{-1}, \tag{58}$$

where, $\mathcal{R}$ is now the resolvent operator for the standard linear system $\mathbf{L}$.

The asymptotic terms can now be examined sequentially noting that by construction the extended variables vanish for the non-linear term, which has the form $\widehat{\mathcal{N}'} = [\mathcal{N}'; \mathbf{0}; \mathbf{0}]$, and hence so do all $\widehat{\mathbf{h}}_{i_1,\ldots,i_k} = [\mathbf{h}_{i_1,\ldots,i_k}; \mathbf{0}; \mathbf{0}]$ derived from it. Starting at $\mathcal{O}(|\mathbf{z}|^2)$ and $(i_1, i_2) = (1, 1)$, one has the resolvent equation in the extended system,

$$[\widehat{\mathcal{R}}(\omega_{1,1})]^{-1}\widehat{\mathbf{y}}_{1,1} + \sum_{j=1}^{m} g_{1,1}^j\widehat{\mathbf{v}}_j = \widehat{\mathbf{h}}_{1,1} \tag{59}$$

The extended variables vanish for $\widehat{\mathbf{h}}_{1,1}$ and therefore, regardless of resonance, they also vanish for $\widehat{\mathbf{y}}_{1,1}$, which then has the form $\widehat{\mathbf{y}}_{1,1} = [\mathbf{y}_{1,1}; \mathbf{0}; \mathbf{0}]$. Hence equation (59) reduces to,

$$[\mathcal{R}(\omega_{1,1})]^{-1}\mathbf{y}_{1,1} + \sum_{j=1}^{n} g_{1,1}^j\mathbf{v}_j = \mathbf{h}_{1,1}, \tag{60}$$

with $g_{1,1}^j = 0$, for $j \in \{n+1, \ldots, m\}$.

For the next term $(i_1, i_2) = (1, 2)$, the right hand side now also contains contributions $\widehat{\mathbf{h}}_{1,2}^O$ due to the off-diagonal terms of $\widehat{\mathbf{K}}$. These however can only come from the preceeding term, $\widehat{\mathbf{y}}_{1,1}$, and therefore off-diagonal contributions also have the form $\widehat{\mathbf{h}}_{1,2}^O = [\mathbf{h}_{1,2}^O; \mathbf{0}; \mathbf{0}]$, which further implies that $\widehat{\mathbf{y}}_{1,2} = [\mathbf{y}_{1,2}; \mathbf{0}; \mathbf{0}]$. The resulting equation being,

$$[\mathcal{R}(\omega_{1,2})]^{-1}\mathbf{y}_{1,2} + \sum_{j=1}^{n} g_{1,2}^j\mathbf{v}_j = \mathbf{h}_{1,2} - \mathbf{h}_{1,2}^O, \tag{61}$$

with $g_{1,2}^j = 0$, for $j \in \{n+1, \ldots, m\}$.

This logic repeats for all $(i_1, i_2) \in \boldsymbol{\mathcal{I}}\{(2), [1, m]\}$ and all unknown vectors at second order have the form $\widehat{\mathbf{y}}_{i_1,i_2} = [\mathbf{y}_{i_1,i_2}; \mathbf{0}; \mathbf{0}]$ and therefore the resolvent equations can be solved using the original linear system $\mathbf{L}$.

At $\mathcal{O}(|\mathbf{z}|^3)$ the right hand side also has contributions from $\widehat{\mathbf{h}}_{i_1,i_2,i_3}^L$ that arise from the lower order asymptotic terms. However, these arise only due to the second order asymptotic terms $\widehat{\mathbf{y}}_{i_1,i_2}$ and therefore, $\widehat{\mathbf{h}}_{i_1,i_2,i_3}^L = [\mathbf{h}_{i_1,i_2,i_3}^L; \mathbf{0}; \mathbf{0}]$. The same logical arguments presented earlier apply and hence all third order asymptotic terms also have the form $\widehat{\mathbf{y}}_{i_1,i_2,i_3} = [\mathbf{y}_{i_1,i_2,i_3}; \mathbf{0}; \mathbf{0}]$ and only require the standard linear system $\mathbf{L}$ for their evaluation. As noted earlier, equations for the higher order asymptotic terms have the same essential structure as the third-order equations and hence at all orders the asymptotic terms have the form $\widehat{\mathbf{y}}_{i_1,\ldots,i_k} = [\mathbf{y}_{i_1,\ldots,i_k}; \mathbf{0}; \mathbf{0}]$ and only require the standard linear system $\mathbf{L}$ for their evaluation.

Equations (54) and (56) may be rewritten for the standard system as,

$$[\mathcal{R}(\omega_{i_1,\ldots,i_k})]^{-1}\mathbf{y}_{i_1,\ldots,i_k} + \sum_{j=1}^{n} g_{i_1,\ldots,i_k}^j\mathbf{v}_j = \mathbf{h}_{i_1,\ldots,i_k} - \mathbf{h}_{i_1,\ldots,i_k}^O - \mathbf{h}_{i_1,\ldots,i_k}^L, \tag{62a}$$

$$\omega_{i_1,\ldots,i_k} = \widehat{K}_{i_1,i_1} + \widehat{K}_{i_2,i_2} + \ldots + \widehat{K}_{i_k,i_k}. \tag{62b}$$

with the resonance condition being the same, *i.e.* ,

$$\omega_{i_1,\ldots,i_k} = \widehat{K}_{j,j}, \qquad j \in 1,\ldots,m, \tag{63}$$

which then defines the normal form coefficients for $j \in \{1,\ldots,n\}$,

$$g^j_{i_1,\ldots,i_k} = \begin{cases} \mathbf{w}^\dagger_j(\mathbf{h}_{i_1,\ldots,i_k} - \mathbf{h}^L_{i_1,\ldots,i_k} \\ \qquad -\mathbf{h}^O_{i_1,\ldots,i_k}), & \textit{if, } \omega_{i_1,\ldots,i_k} = \widehat{K}_{j,j} \\ 0 = & \textit{otherwise}, \end{cases} \tag{64}$$

and $\mathbf{w}^\dagger_j\mathbf{y}_{i_1,\ldots,i_k} = 0$ in case of resonance. For all $j \in \{n+1,\ldots,m\}$,

$$g^j_{i_1,\ldots,i_k} = 0, \tag{65}$$

regardless of resonance.

The entire procedure for the evaluation of the normal form is presented succinctly in algorithm 3.

Algorithm 3 Center–Manifold: Normal form

Require: System: $\partial\mathbf{u}/\partial t = \mathbf{L}\mathbf{u} + \mathcal{N}(\mathbf{u},\boldsymbol{\mu}^b)$.
Ensure: $\mathcal{N}(\mathbf{0},\boldsymbol{\mu}^b) = \mathbf{0}$; $\partial\mathcal{N}/\partial\mathbf{u} = \mathbf{0}$.

Require: Tangent Space: $\mathbf{W}^\dagger\mathbf{L}\mathbf{V} = \boldsymbol{\Lambda}_C$.
  /* *Center subspace dimension:* n. */

Require: Extension via $\boldsymbol{\nu},\boldsymbol{\theta}$ and $\mathcal{N}'(\mathbf{u},\boldsymbol{\nu},\boldsymbol{\theta})$.
  Evaluate $\widehat{\mathbf{V}},\widehat{\mathbf{K}}$ via algorithm 1.
  /* *Center subspace dimension:* $m = n+l+h$. */

Require: Maximum Asymptotic order: k_{max}
  /* *Substitute asymptotic expansions* */
  for $(k \in 2,3,\ldots,k_{max})$ **do**
    for $(i_1,\ldots,i_k) \in \boldsymbol{\mathcal{I}}\{(k),[1,m]\}$ **do**
      /* *Build all r.h.s. contributions.* */
      $\mathbf{r} = \mathbf{h}_{i_1,\ldots,i_k} - \mathbf{h}^O_{i_1,\ldots,i_k} - \mathbf{h}^L_{i_1,\ldots,i_k}$
      $\omega = \widehat{K}_{i_1,i_1} + \widehat{K}_{i_2,i_2} + \ldots + \widehat{K}_{i_k,i_k}$
      /* *Build* $\mathbf{g}_{i_1,\ldots,i_k}$*, Eq.* (64) */
      for $(j \in 1,\ldots,n)$ **do**
        if $(\omega = \widehat{K}_{j,j})$ **then**
          $g^j_{i_1,\ldots,i_k} = \mathbf{w}^\dagger_j\mathbf{r}$
        else
          $g^j_{i_1,\ldots,i_k} = 0$
        end if
      end for
      for $(j \in n+1,\ldots,m)$ **do**
        $g^j_{i_1,\ldots,i_k} = 0$
      end for
      /* *Build singular components* */
      $\mathbf{r}_s = \sum_{j=1}^n \mathbf{v}_j g^j_{i_1,\ldots,i_k}$
      /* *Evaluate resolvent* */
      $\mathbf{y}_{i_1,\ldots,i_k} = \mathcal{R}(\omega)[\mathbf{r} - \mathbf{r}_s]$ /* *Eq.* (62)
      /* $\mathcal{R}(\omega) = (\omega\mathbf{I} - \mathbf{L})^{-1}$ */
    end for
  end for
  return $\mathcal{G}(\mathbf{z}),\mathcal{Y}(\mathbf{z})$.

3.2 A Graph Style Parameterization

In the earlier subsection it was shown that even though one starts from an extended system, for the evaluation of the higher order asymptotic terms of the center-manifold (in its normal form) only the standard linear system is required and the system extension only affects the tangent (linear) space. This aspect however is not unique to the normal form and other parameterizations can be constructed in which the asymptotic center-manifold of a system can be evaluated using just the standard linear operator $\mathbf{L}$. Here we construct one such parameterization.

Starting again with the asymptotic representations of $\mathcal{G}(\mathbf{z})$ and $\mathcal{Y}(\mathbf{z})$ as in equations (38) and (39) respectively, one may constrain all higher order terms $\widehat{\mathbf{y}}_{i_1,\ldots,i_k}$ to lie in the complement of the center subspace (spanned by $\widehat{\mathbf{V}}$), resulting in,

$$\widehat{\mathbf{w}}^\dagger_j\widehat{\mathbf{y}}_{i_1,\ldots,i_k} = \mathbf{0}, \qquad \forall j \in 1,\ldots,m, \tag{66}$$

for all $(i_1,\ldots,i_k) \in \boldsymbol{\mathcal{I}}\{(k),[1,m]\}$, $k \geq 2$. Note that this condition applies regardless of resonance.

At $\mathcal{O}(|\mathbf{z}|)$ one again obtains equation (42) and, as in the case of the normal form, the unknowns $\widehat{\mathbf{y}}_{i_1},\mathbf{g}_{i_1}$ *etc.*, can be associated with the generalized eigenvalue problem resulting in equation (43). At first order the expressions are identical to those obtained for the normal form, which is unsurprising since the $\mathcal{O}(|\mathbf{z}|)$ terms only represent the center subspace.

Since the linear part of the reduced representation is given by $\widehat{\mathbf{K}}$, one again obtains a lower triangular system of equations for the unknown vectors $\widehat{\mathbf{y}}_{i_1,\ldots,i_k}$ at all asymptotic orders, which can be solved sequentially one vector at a a time. One may therefore represent the general expression for the evaluation of higher order terms as,

$$[\widehat{\mathcal{R}}(\omega_{i_1,\ldots,i_k})]^{-1}\widehat{\mathbf{y}}_{i_1,\ldots,i_k} + \sum_{j=1}^{m} g^j_{i_1,\ldots,i_k}\widehat{\mathbf{v}}_j = \widehat{\mathbf{h}}_{i_1,\ldots,i_k} - \widehat{\mathbf{h}}^O_{i_1,\ldots,i_k} - \widehat{\mathbf{h}}^L_{i_1,\ldots,i_k}. \tag{67a}$$

$$\omega_{i_1,\ldots,i_k} = \widehat{K}_{i_1,i_1} + \widehat{K}_{i_2,i_2} + \ldots + \widehat{K}_{i_k,i_k}. \tag{67b}$$

Since the parameterization is defined by equation (66), one obtains the definition of the asymptotic terms of $\mathcal{G}$ as,

$$g^j_{i_1,\ldots,i_k} = \widehat{\mathbf{w}}^\dagger_j(\widehat{\mathbf{h}}_{i_1,\ldots,i_k} - \widehat{\mathbf{h}}^O_{i_1,\ldots,i_k} - \widehat{\mathbf{h}}^L_{i_1,\ldots,i_k}), \tag{68}$$

$\forall j \in 1,\ldots,m$.

The extended variables vanish for $\widehat{\mathbf{h}}_{i_1,\ldots,i_k}$ by construction and, following the logical arguments presented in the previous subsection, it may be verified that the extended variables also vanish for $\widehat{\mathbf{h}}^O_{i_1,\ldots,i_k}, \widehat{\mathbf{h}}^L_{i_1,\ldots,i_k}$ and $\widehat{\mathbf{y}}_{i_1,\ldots,i_k}$ (with $k \geq 2$). This then implies that,

$$g^j_{i_1,\ldots,i_k} = 0, \qquad \forall j \in n+1,\ldots,m. \tag{69}$$

Equation (67) then reduces to its single component form,

$$[\mathcal{R}(\omega_{i_1,\ldots,i_k})]^{-1}\mathbf{y}_{i_1,\ldots,i_k} + \sum_{j=1}^{n} g^j_{i_1,\ldots,i_k}\mathbf{v}_j = \mathbf{h}_{i_1,\ldots,i_k} - \mathbf{h}^O_{i_1,\ldots,i_k} - \mathbf{h}^L_{i_1,\ldots,i_k}. \tag{70a}$$

$$\omega_{i_1,\ldots,i_k} = \widehat{K}_{i_1,i_1} + \widehat{K}_{i_2,i_2} + \ldots + \widehat{K}_{i_k,i_k}. \tag{70b}$$

One may define the projection operator on to the complement of the center subspace of $\mathbf{L}$ as,

$$\mathbf{Q} = \mathbf{I} - \mathbf{V}\mathbf{W}^\dagger. \tag{71}$$

By construction we have, $\mathbf{Q}\mathbf{y}_{i_1,\ldots,i_k} = \mathbf{y}_{i_1,\ldots,i_k}$ and, $\mathbf{Q}\sum_{j=1}^{n} g^j_{i_1,\ldots,i_k}\mathbf{v}_j = \mathbf{0}$. Expressing the resolvent operator restricted to the complement of the center subspace as

$$\mathcal{R}_s(\omega) = [\mathbf{Q}(\omega\mathbf{I} - \mathbf{L})\mathbf{Q}]^{-1} \tag{72}$$

Equation (70) can then be rewritten in a restricted subspace as,

$$[\mathcal{R}_s(\omega_{i_1,\ldots,i_k})]^{-1}\mathbf{y}_{i_1,\ldots,i_k} = \mathbf{Q}\left[\mathbf{h}_{i_1,\ldots,i_k} - \mathbf{h}^O_{i_1,\ldots,i_k} - \mathbf{h}^L_{i_1,\ldots,i_k}\right], \tag{73a}$$

$$\omega_{i_1,\ldots,i_k} = \widehat{K}_{i_1,i_1} + \widehat{K}_{i_2,i_2} + \ldots + \widehat{K}_{i_k,i_k}. \tag{73b}$$

with the asymptotic components of $\mathcal{G}$ defined as

$$g^j_{i_1,\ldots,i_k} = \mathbf{w}^\dagger_j(\mathbf{h}_{i_1,\ldots,i_k} - \mathbf{h}^O_{i_1,\ldots,i_k} - \mathbf{h}^L_{i_1,\ldots,i_k}), \quad \textit{for } j \in 1,\ldots,n, \tag{74a}$$

$$g^j_{i_1,\ldots,i_k} = 0, \quad \textit{for } j \in n+1,\ldots,m. \tag{74b}$$

This particular graph style parameterization has been utilized by Negi [9] for the Navier–Stokes system where the Reynolds number perturbations have been used for the system extension. The procedure for obtaining the graph representation is summarized in algorithm 4.

The resulting expressions for the normal form given by equations (62), (64) and, the graph style parameterization given by equations (73), (74) are quite similar. The similarity is due to the fact that the tangent space in both cases is identical, with $\widehat{\mathbf{K}}$ representing the linear part of the parameterization $\mathcal{G}(\mathbf{z})$. Typically one is interested in the normal form of the reduced equations since they result in the "simplest form" of the representation. The particular graph parameterization however is advantageous when one needs to consider the mapping from the full system state to the reduced manifold, *i.e.* , $\widehat{\mathbf{u}} \mapsto \mathbf{z}$. For both normal form and the graph parameterization, mapping that is explicitly constructed is $\mathcal{Y} : \mathbf{z} \mapsto \widehat{\mathbf{u}}$. The inverse map $\mathcal{Y}^{-1} : \widehat{\mathbf{u}} \mapsto \mathbf{z}$ is not constructed. It is also difficult to construct from the knowledge of $\mathcal{Y}$ itself because $\mathcal{Y}$ is constructed asymptotically and is hardly ever evaluated beyond a few orders. However, since the graph parameterization is constructed with the property that higher order terms lie in the complement of the tangent space, the inverse map $\mathcal{Y}^{-1}$ can be easily constructed by leveraging this property. Therefore, recalling that the first order vectors $\mathbf{y}_i = \widehat{\mathbf{v}}_i$, we have,

$$\widehat{\mathbf{u}} = \mathcal{Y}(\mathbf{z}) = \sum_{(1)}^{[1,m]} \widehat{\mathbf{y}}_{i_1} z_{i_1} + \sum_{(2)}^{[1,m]} \widehat{\mathbf{y}}_{i_1,i_2} z_{i_1} z_{i_2} + \sum_{(3)}^{[1,m]} \widehat{\mathbf{y}}_{i_1,i_2,i_3} z_{i_1} z_{i_2} z_{i_3} + \ldots,$$

Algorithm 4 Center–Manifold: Graph parameterization

Require: System: $\partial \mathbf{u}/\partial t = \mathbf{L}\mathbf{u} + \mathcal{N}(\mathbf{u}, \boldsymbol{\mu}^b)$.
Ensure: $\mathcal{N}(\mathbf{0}, \boldsymbol{\mu}^b) = \mathbf{0}$; $\partial \mathcal{N}/\partial \mathbf{u} = \mathbf{0}$.

Require: Tangent Space: $\mathbf{W}^\dagger \mathbf{L}\mathbf{V} = \boldsymbol{\Lambda}_C$.
  /* *Center subspace dimension: n.* */

Require: Extension via $\boldsymbol{\nu}, \boldsymbol{\theta}$ and $\mathcal{N}'(\mathbf{u}, \boldsymbol{\nu}, \boldsymbol{\theta})$.
  Evaluate $\widehat{\mathbf{V}}, \widehat{\mathbf{K}}$ via algorithm 1.
  /* *Center subspace dimension: $m = n+l+h$.* */
  $\mathbf{Q} = \mathbf{I} - \mathbf{V}\mathbf{W}^\dagger$ /* *Subspace projector.*

Require: Maximum Asymptotic order: k_{max}
  for $(k \in 2, 3, \ldots, k_{max})$ **do**
    for $(i_1, \ldots, i_k) \in \boldsymbol{\mathcal{I}}\{(k), [1, m]\}$ **do**
      /* *Build all r.h.s. contributions.* */
      $\mathbf{r} = \mathbf{h}_{i_1,\ldots,i_k} - \mathbf{h}^O_{i_1,\ldots,i_k} - \mathbf{h}^L_{i_1,\ldots,i_k}$
      $\omega = \widehat{K}_{i_1,i_1} + \widehat{K}_{i_2,i_2} + \ldots + \widehat{K}_{i_k,i_k}$
      /* *Build $\mathbf{g}_{i_1,\ldots,i_k}$, Eq. (74)* */
      for $(j \in 1, \ldots, n)$ **do**
        $g^j_{i_1,\ldots,i_k} = \mathbf{w}^\dagger_j \mathbf{r}$
      end for
      for $(j \in n+1, \ldots, m)$ **do**
        $g^j_{i_1,\ldots,i_k} = 0$
      end for
      /* *Evaluate restricted resolvent* */
      $\mathbf{y}_{i_1,\ldots,i_k} = \mathcal{R}_s(\omega)\mathbf{Q}\mathbf{r}$, /* *Eq. (73)*
      with, $\mathbf{Q}\mathbf{y}_{i_1,\ldots,i_k} = \mathbf{0}$.
      /* *$\mathcal{R}_s(\omega) = [\mathbf{Q}(\omega\mathbf{I} - \mathbf{L})\mathbf{Q}]^{-1}$* */
    end for
  end for
  return $\mathcal{G}(\mathbf{z}), \mathcal{Y}(\mathbf{z})$.

$$\begin{aligned}\implies \widehat{\mathbf{W}}^\dagger \hat{\mathbf{u}} &= \sum_{(1)}^{[1,m]} \widehat{\mathbf{W}}^\dagger \hat{\mathbf{v}}_{i_1} z_{i_1} + \sum_{(2)}^{[1,m]} \widehat{\mathbf{W}}^\dagger \hat{\mathbf{y}}_{i_1,i_2} z_{i_1} z_{i_2} \\ &\quad + \sum_{(3)}^{[1,m]} \widehat{\mathbf{W}}^\dagger \hat{\mathbf{y}}_{i_1,i_2,i_3} z_{i_1} z_{i_2} z_{i_3} + \ldots, \\ \implies \widehat{\mathbf{W}}^\dagger \hat{\mathbf{u}} &= \hat{\mathbf{I}}\mathbf{z} + \mathbf{0} + \mathbf{0} + \ldots.\end{aligned}$$

Therefore, when considering the graph parameterization presented earlier, the inverse map $\mathcal{Y}^{-1}$: $\hat{\mathbf{u}} \mapsto \mathbf{z}$ is simply,

$$\mathbf{z} = \mathcal{Y}^{-1}(\hat{\mathbf{u}}) = \widehat{\mathbf{W}}^\dagger \hat{\mathbf{u}}. \tag{75}$$

For the normal form, one does not get this simple inverse map since the higher order terms generically do not lie in the complement of the tangent space $(\widehat{\mathbf{V}})$.

4 A Model Problem

The asymptotic center-manifold derived in section 3 is applied to a Ginzburg-Landau model described by a complex-conjugate pair of equations as,

$$\frac{\partial A}{\partial t} = \left[\delta_1 \nabla + \delta_2 + \delta_3 x + \delta_4 \nabla^2\right] A + \delta_5 |A|^2 A, \tag{76a}$$

$$\frac{\partial A^*}{\partial t} = \left[\delta_1^* \nabla + \delta_2^* + \delta_3^* x + \delta_4^* \nabla^2\right] A^* + \delta_5^* |A|^2 A^*. \tag{76b}$$

Where, x is the spatial coordinate, ∇ and ∇^2 are the one-dimensional gradient and Laplacian operators, and the superscript * represents complex conjugation. Typically such a model would arise as the simplified dynamics of a more detailed dynamical system. Here the Ginzburg-Landau model itself is treated as the underlying dynamical model, which represents the standard system in our context. Such models have often been utilized to investigate local and global bifurcations in spatially developing flows [12, 13]. It is easily seen that $A(x) = 0$ represents a fixed point of the system, with the linear operators given by

$$\begin{aligned}\mathcal{L}_0 &= \left[\delta_1 \nabla + \delta_2 + \delta_3 x + \delta_4 \nabla^2\right], \\ \mathcal{L}_0^* &= \left[\delta_1^* \nabla + \delta_2^* + \delta_3^* x + \delta_4^* \nabla^2\right].\end{aligned}$$

The linear operator for both A and A^* are written together as

$$\mathbf{L} = \begin{bmatrix} \mathcal{L}_0 & 0 \\ 0 & \mathcal{L}_0^* \end{bmatrix}, \tag{77}$$

which represents the linear operator for the standard system. The state variable is denoted by $\mathbf{u} = [A; A^*]$.

For semi-infinite domains, $x \in [0, \infty)$, the spectrum for $\mathcal{L}_0$ is known analytically [12] and is given by,

$$\omega_n = \delta_2 - \frac{\delta_1^2}{4\delta_4} + (\delta_4 \delta_3^2)^{1/3} \zeta_n, \tag{78}$$

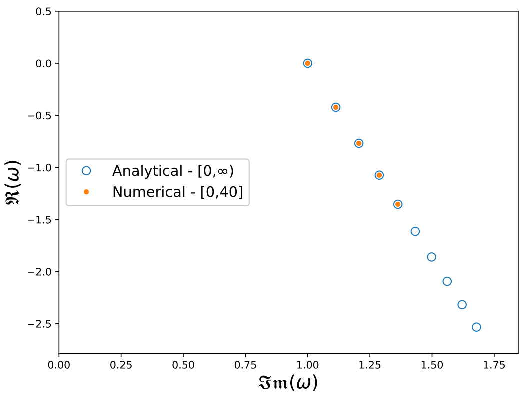

Fig. 3: Comparison of the numerically evaluated spectrum of the Ginzburg-Landau model ($\mathcal{L}_0$) in the truncated domain with the analytical spectrum in the semi-infinite domain.

and the corrresponding complex conjugate for $\mathcal{L}_0^*$. Here ζ_n denotes the countable set of zeros of the Airy function. The system parameters are chosen *ad hoc* as $\delta_1^b = -1.0$, $\delta_2^b = 0.741 + 1.025i$, $\delta_3^b = -0.125$, $\delta_4^b = (1.0 - i)/\sqrt{2}$ and $\delta_5^b = -0.1 + 0.1i$, such that the system is at the bifurcation point with a single critical eigenmode with eigenvalue $\omega_1 = 1.0i$ for $\mathcal{L}_0$ and its conjugate, $-1.0i$ for $\mathcal{L}_0^*$, and all other eigenvalues have negative real parts.

A center-manifold approximation of the model problem requires the evaluation of restricted resolvents, which is a difficult problem to solve analytically. Therefore the problem is considered in a trucated domain of $x \in [0, 40]$ with the boundary conditions of $A = 0$ at $x = 0$ and, $\partial A/\partial x = 0$ (natural boundary conditions) at $x = 40$. Due to the spatially localized nature of the first few eigenvectors (which dictate the global dynamics), this truncation does not affect the dynamics in any significant manner, as can be seen from the comparison of the numerically evaluated spectrum (via the implicitly restarted Arnoldi method [14]) in the truncated domain to the analytical spectrum in the semi-infinite domain (figure 3).

The standard system then has the critical eigenvalue and eigenvector matrices as

$$\mathbf{\Lambda}_C = \begin{bmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{bmatrix}; \quad \mathbf{V} = [\mathbf{v}_1; \mathbf{v}_2]; \quad \mathbf{W} = [\mathbf{w}_1; \mathbf{w}_2],$$

where the eigenvectors have the following form in terms of the two components,

$$\begin{aligned} \mathbf{v}_1 &= [\phi_c; \mathbf{0}]; & \mathbf{v}_2 &= [\mathbf{0}; \phi_c^*], \\ \mathbf{w}_1 &= [\chi_c; \mathbf{0}]; & \mathbf{w}_2 &= [\mathbf{0}; \chi_c^*], \end{aligned}$$

The numerically calculated eigenvectors ϕ_c and χ_c are shown in figure 4.

The system in equation (76) has a total of five parameters (ten including the complex-conjugate pairs) that can be varied from their bifurcation values. In order to avoid an explosion in the number of terms T_m^k arising from the combinatorial problem, only the variations in δ_4 and δ_4^* are considered, which suffice to illustrate the construction of the asymptotic center-manifold under parameter variations. Although the parameters are complex-conjugate pairs, it turns out to be simplest to treat them all as independent parameters and enforce the complex-conjugate property later when parameter values are prescribed, which is essentially the same as treating A and A^* independently and having two independent critical eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$. The extended variable for the parameter evolution is then $\boldsymbol{\nu} = [\delta_4'; \delta_4'^*]$, with

$$\delta_4 = \delta_4^b + \delta_4', \qquad \delta_4^* = (\delta_4^b)^* + \delta_4'^*.$$

We consider the harmonically forced system with, $h = 5$ and angular frequencies, $\boldsymbol{\lambda}^H = [0.0; 1i; -1.0i; 2.37i; -2.37i]$. The external forcing is given by equation (8), and we specify the forcing shapes $\boldsymbol{f}_i^H$, *ad hoc*, based on,

$$\xi(x) = \frac{1}{\sqrt{\pi}} e^{-(x-7.0)^2}$$

with the forcing shapes being,

$$\boldsymbol{f}_1^H = \frac{1}{\sqrt{2}} \begin{bmatrix} \xi(x) \\ \xi(x) \end{bmatrix}, \boldsymbol{f}_2^H = \begin{bmatrix} \xi(x) \\ \mathbf{0} \end{bmatrix}, \quad \boldsymbol{f}_3^H = \begin{bmatrix} \mathbf{0} \\ \xi(x) \end{bmatrix},$$
$$\boldsymbol{f}_4^H = \begin{bmatrix} \xi(x) \\ \mathbf{0} \end{bmatrix}, \boldsymbol{f}_5^H = \begin{bmatrix} \mathbf{0} \\ \xi(x) \end{bmatrix}.$$

The extended state vector is denoted as $\widehat{A} = [A; A^*; \boldsymbol{\nu}]$. With the knowledge of the linear operator and having calculated the critical eigenmodes, all the expressions for the center-manifold approximation of the extended system can now be easily built. The extended linear operator is given by,

$$\widehat{\mathcal{L}} = \begin{bmatrix} \mathcal{L} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \mathcal{L}_0 & 0 & 0 \\ 0 & \mathcal{L}_0^* & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

This results in six critical modes for the extended system - two critical modes resulting from the original linear system $\mathcal{L}$ and four parameter modes due to system extension. The direct and adjoint eigenvector matrices are given by

$$\begin{aligned} \widehat{\mathbf{V}} &= [\widehat{\boldsymbol{v}}_1; \widehat{\boldsymbol{v}}_2; \widehat{\boldsymbol{v}}_3; \widehat{\boldsymbol{v}}_4; \widehat{\boldsymbol{v}}_5; \widehat{\boldsymbol{v}}_6] = \begin{bmatrix} 0 & 0 & 0 & 0 & \phi_c & 0 \\ 0 & 0 & 0 & 0 & 0 & \phi_c^* \\ \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 & \boldsymbol{e}_4 & 0 & 0 \end{bmatrix} \\ \widehat{\mathbf{W}} &= [\widehat{\boldsymbol{w}}_1; \widehat{\boldsymbol{w}}_2; \widehat{\boldsymbol{w}}_3; \widehat{\boldsymbol{w}}_4; \widehat{\boldsymbol{w}}_5; \widehat{\boldsymbol{w}}_6] = \begin{bmatrix} 0 & 0 & 0 & 0 & \chi_c & 0 \\ 0 & 0 & 0 & 0 & 0 & \chi_c^* \\ \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 & \boldsymbol{e}_4 & 0 & 0 \end{bmatrix}. \end{aligned}$$

resulting in the reduced system matrix,

$$\widehat{\mathbf{K}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \omega_c & 0 \\ 0 & 0 & 0 & 0 & 0 & \omega_c^* \end{bmatrix}.$$

Due to the absence of purely real critical eigenvalues in the original system, the $\mathbf{\Gamma}$ matrix vanishes.

To avoid confusion with the spatial variable x, the critical mode amplitudes are denoted by $\mathbf{z} = [z_1; z_2; z_3; z_4; z_5; z_6]$, the stable component solution of the original system is given by $A_s = [A_{1s}; A_{2s}]$, and in the extended space it is $\widehat{A}_s = [A_{1s}; A_{2s}; 0]$. Thus the total solution is written as,

$$\widehat{A} = \widehat{\mathbf{V}}\mathbf{z} + \widehat{A}_s.$$

The non-linear term of the original system is given by $\mathcal{N}_{NL} = [\mathcal{N}_1; \mathcal{N}_2]$, with $\mathcal{N}_1$ and $\mathcal{N}_2$ defined as,

$$\begin{aligned} \mathcal{N}_1 =& z_1 z_5 x \phi_c + z_1 x A_{1s} + z_2 z_5 \nabla^2 \phi_c + z_2 \nabla^2 A_{1s} \\ &+ \delta_5^b (\phi_c^* z_6 + A_{2s})(\phi_c z_5 + A_{1s})(\phi_c z_5 + A_{1s}), \\ \mathcal{N}_2 =& z_3 z_6 x \phi_c^* + z_3 x A_{2s} + z_4 z_6 \nabla^2 \phi_c^* + z_4 \nabla^2 A_{2s} \\ &+ \delta_5^{b*} (\phi_c z_5 + A_{1s})(\phi_c^* z_6 + A_{2s})(\phi_c^* z_6 + A_{2s}), \end{aligned}$$

and in the extended space it is defined as $\widehat{\mathcal{N}} = [\mathcal{N}_1; \mathcal{N}_2; 0]$. The evolution equations for the critical and stable subspaces are

$$\begin{aligned} \frac{dz_i}{dt} =& 0, \qquad \text{for } i \in 1,2,3,4, && \text{(79a)} \\ \frac{dz_5}{dt} =& \omega_c z_5 + \chi_c^H \mathcal{N}_1, && \text{(79b)} \\ \frac{dz_6}{dt} =& \omega_c^* z_6 + \chi_c^{*H} \mathcal{N}_2, && \text{(79c)} \\ \frac{\partial A_s}{\partial t} =& \mathbf{Q}\mathcal{L}\mathbf{Q}A_s + \mathbf{Q}\mathcal{N}. && \text{(79d)} \end{aligned}$$

The center-manifold theorem is applied to equation (79) and the stable component solution is evaluated asympotically as a power series in $\mathbf{z}$,

$$\begin{aligned} \mathfrak{h}(\mathbf{z}) &= \sum_{\alpha=1}^{6} \sum_{\beta=\alpha}^{6} z_\alpha z_\beta \boldsymbol{y}_{\alpha,\beta} + \mathcal{O}(\mathbf{z}^3), \\ \boldsymbol{y}_{\alpha,\beta} &= \begin{Bmatrix} \boldsymbol{y}_{\alpha,\beta}^1 \\ \boldsymbol{y}_{\alpha,\beta}^2 \end{Bmatrix}. \end{aligned}$$

For $M = 6$, a total of $T_2 = 21$ terms are obtained at second order for the asymptotic approximation. All coupling terms $\boldsymbol{g}_{\alpha,\beta}^c$ vanish identically and only the following nontrivial forcing terms $\boldsymbol{g}_{\alpha,\beta}$ are obtained at second order,

$$\begin{aligned} \boldsymbol{g}_{1,5} =& \mathbf{Q} \cdot x\mathbf{v}_5 = \begin{Bmatrix} \mathbf{Q}_0 x \phi_c \\ 0 \end{Bmatrix}, \\ \boldsymbol{g}_{2,5} =& \mathbf{Q} \cdot \nabla^2 \mathbf{v}_5 = \begin{Bmatrix} \mathbf{Q}_0 \nabla^2 \phi_c \\ 0 \end{Bmatrix}, \\ \boldsymbol{g}_{3,6} =& \mathbf{Q} \cdot x\mathbf{v}_6 = \begin{Bmatrix} 0 \\ \mathbf{Q}_0^* x \phi_c^* \end{Bmatrix}, \\ \boldsymbol{g}_{4,6} =& \mathbf{Q} \cdot \nabla^2 \mathbf{v}_6 = \begin{Bmatrix} 0 \\ \mathbf{Q}_0^* \nabla^2 \phi_c^* \end{Bmatrix}, \end{aligned}$$

with all other forcing terms $\boldsymbol{g}_{\alpha,\beta}$ vanishing identically at second order. This leads to the restricted resolvent solutions $\mathbf{y}_{\alpha,\beta}$ that satisfy,

$$\begin{aligned} \mathbf{Q}[(0+\omega_c)\mathbf{I} - \mathcal{L}]\mathbf{Q}\mathbf{y}_{1,5} =& \boldsymbol{g}_{1,5}, && \text{(80a)} \\ \mathbf{Q}[(0+\omega_c)\mathbf{I} - \mathcal{L}]\mathbf{Q}\mathbf{y}_{2,5} =& \boldsymbol{g}_{2,5}, && \text{(80b)} \\ \mathbf{Q}[(0+\omega_c^*)\mathbf{I} - \mathcal{L}]\mathbf{Q}\mathbf{y}_{3,6} =& \boldsymbol{g}_{3,6}, && \text{(80c)} \\ \mathbf{Q}[(0+\omega_c^*)\mathbf{I} - \mathcal{L}]\mathbf{Q}\mathbf{y}_{4,6} =& \boldsymbol{g}_{4,6}, && \text{(80d)} \end{aligned}$$

with all other second order terms $\mathbf{y}_{\alpha,\beta}$ vanishing. The resolvent solutions in terms of the individual

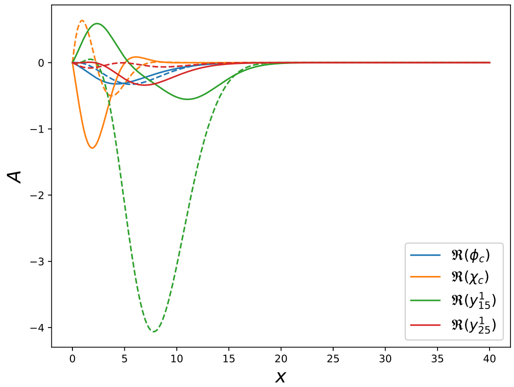

Fig. 4: Numerically evaluated eigenvector fields ϕ_c and χ_c, along with the second-order approximations of stable components $\mathbf{y}_{1,5}$ and $\mathbf{y}_{2,5}$. The real pars of the fields depicted by solid lines while the imaginary parts are depicted with dashed lines of the same color.

fields are written as

$$\mathbf{Q}_0[(0+\omega_c)\mathbf{I}-\mathcal{L}_0]\mathbf{Q}_0\mathbf{y}^1_{1,5} = \mathbf{Q}_0 x\phi_c, \tag{81a}$$
$$\mathbf{Q}_0[(0+\omega_c)\mathbf{I}-\mathcal{L}_0]\mathbf{Q}_0\mathbf{y}^1_{2,5} = \mathbf{Q}_0 \nabla^2\phi_c, \tag{81b}$$
$$\mathbf{Q}^*_0[(0+\omega^*_c)\mathbf{I}-\mathcal{L}^*_0]\mathbf{Q}^*_0\mathbf{y}^2_{3,6} = \mathbf{Q}^*_0 x\phi^*_c, \tag{81c}$$
$$\mathbf{Q}^*_0[(0+\omega^*_c)\mathbf{I}-\mathcal{L}^*_0]\mathbf{Q}^*_0\mathbf{y}^2_{2,5} = \mathbf{Q}^*_0 \nabla^2\phi^*_c. \tag{81d}$$

Obviously, $\mathbf{y}^2_{1,5} = \mathbf{y}^2_{2,5} = \mathbf{y}^1_{3,6} = \mathbf{y}^1_{4,6} = 0$, with $\mathbf{y}^2_{3,6} = (\mathbf{y}^1_{1,5})^*$ and $\mathbf{y}^2_{4,6} = (\mathbf{y}^1_{2,5})^*$, as would be expected from the structure of the original equations. The second-order asymptotic fields $\mathbf{y}^1_{1,5}$ and $\mathbf{y}^1_{2,5}$ (along with the critical eigenvectors) are depicted in figure 4. The fields $\phi^*_c, \chi^*_c, \mathbf{y}^2_{3,6}$ and $\mathbf{y}^2_{4,6}$ are the corresponding complex-conjugates (not shown).

The complex-conjugation property may now be enforced so that $z_6 = z^*_5, z_3 = z^*_1, z_4 = z^*_2$ and, the original parameter perturbations reintroduced, $z_1 = \delta'_3, z_2 = \delta'_4$. The resulting critical mode amplitude equation is

$$\begin{aligned}\frac{dz_5}{dt} =& \omega_c z_5 + \delta'_3(\chi^H_c x\phi_c)z_5 + (\delta'_3)^2(\chi^H_c x\mathbf{y}_{1,5})z_5 \\ &+ \delta'_3\delta'_4(\chi^H_c x\mathbf{y}_{2,5})z_5 + \delta'_4(\chi^H_c\nabla^2\phi_c)z_5 \\ &+ \delta'_3\delta'_4(\chi^H_c\nabla^2\mathbf{y}_{1,5})z_5 + (\delta'_4)^2(\chi^H_c\nabla^2\mathbf{y}_{2,5})z_5 \\ &+ \delta^b_5(\chi^H_c|\phi_c|^2\phi_c)|z_5|^2 z_5,\end{aligned} \tag{82}$$

and its complex conjugate for z_6, mirroring the original structure of equation (76). In this case, the Ginzburg-Landau equation has been reduced to a Stuart-Landau like equation in the vicinity of the bifurcation point. The inner-products in equation (82) are evaluated numerically as,

$$\begin{aligned}\chi^H_c x\phi_c =& \quad 3.011 - 0.807i,\\ \chi^H_c x\mathbf{y}_{1,5} =& \quad 4.015 - 1.076i,\\ \chi^H_c x\mathbf{y}_{2,5} =& \quad -0.036 - 0.582i,\\ \chi^H_c\nabla^2\phi_c =& \quad -0.169 + 0.153i,\\ \chi^H_c\nabla^2\mathbf{y}_{1,5} =& \quad 0.936 + 1.101i,\\ \chi^H_c\nabla^2\mathbf{y}_{2,5} =& \quad 0.194 - 0.114i,\\ \chi^H_c|\phi_c|^2\phi_c =& \quad 0.111 - 0.032i.\end{aligned}$$

The center-manifold reduction obtained in equation (82) can be compared with the original system by evaluating the saturated angular frequencies, Ω, obtained for the full system and its reduced counter-part for parameter perturbations. The comparison is shown in figure 5 for the variation of δ'_3 and δ'_4. The agreement is very good close to the bifurcation point with a systematic deviation arising at larger parameter variations. Such deviation would be expected given the asymptotic nature of the approximation. The deviation is also larger for δ'_3 compared to δ'_4, which could be qualitatively understood apriori, given the relatively large norm of $\mathbf{y}_{1,5}$, as seen in figure 4, which arises due to δ'_3 perturbations.

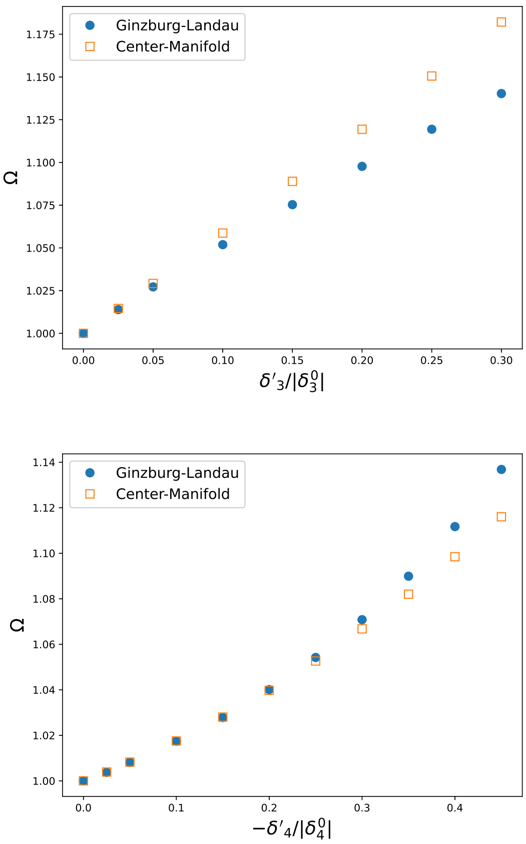

Fig. 5: Comparison of the angular frequencies obtained from the Ginzburg-Landau system, equation (76), and its center-manifold reduction, equation (82) for different values of parameter perturbations. (Top) $\delta'_4 = 0$ with varying δ'_3 and, (bottom) $\delta'_3 = 0$ with varying δ'_4. Note the negative sign in the $x-$axis label in the bottom figure.

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